

***Revised  
Generation Interconnection  
System Impact Study Report***

***For***

***PJM Generation Interconnection Request  
Queue Position AD1-113***

***“Raritan River 230 kV”***

***35 MWE, 75 MWC Uprate***

***December 2018  
Revision 1***

## Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

## Changes from Feasibility Study:

Per request of the Interconnection Customer, the increase in Maximum Facility Output (MFO)/Energy was reduced from 75 MW to 35 MW for the AD1-113 project. PJM determined this change was not a material modification. The Capacity value request did not change. See the updated values in the table below in the “General” section.

## Update to September System Impact Study

The System Impact Study Report has been updated to remove an incorrect statement regarding an overload of the generator’s lead line. Updated ratings have been provided by the Interconnection Customer and incorporated into the model.

The reactive deficiency table was also updated to reflect a 69.4 MVAR lagging deficiency. The customer has indicated that they will cure this deficiency on their side of the Point of Interconnection. Schedule F of the ISA will address the need to cure this deficiency and the consequences if it is not cured.

## General

**CPV Shore, LLC**, the Interconnection Customer (IC), has proposed an increase in output of their existing Woodbridge Energy Center natural gas generating facility located at 1070 Riverside Drive, Keasbey, New Jersey (Attachment 1). The uprate will increase the maximum facility output (MFO) of the facility by **35 MW**, for a total plant MFO of 820 MW. The uprate

will also increase the Capacity value by **75 MW**, for a total of 800 MW being recognized by PJM as Capacity. (See the summary table below.) The proposed in-service date for this project is **December 1, 2018**. **This study does not imply a Jersey Central Power and Light (JCP&L) commitment to this in-service date.**

| Queue Number | MFO (MW) and Winter Energy | Capacity (MW) and Summer Energy |
|--------------|----------------------------|---------------------------------|
| W4-009       | 725                        | 725                             |
| X4-005       | 785 (60 MW increase)       | 725 (0 MW increase)             |
| AD1-113      | 820 (35 MW increase)       | 800 (75 MW increase)            |
| <b>Total</b> | <b>820</b>                 | <b>800</b>                      |

PJM studied the proposed request and evaluated it for compliance with reliability criteria for summer peak conditions in 2021.

## Point of Interconnection

**AD1-113 “Raritan River 230 kV”** uprate project will be an increase to the existing Woodbridge Energy Center natural gas generation plant which interconnects to the JCP&L transmission system at the Raritan River 230 kV bus. As defined by CPV Shore, LLC and shown on Attachment 2, the connection point for this project will be accomplished by utilizing the existing 230 kV point of interconnection for the Woodbridge generation plant at JCP&L’s Raritan River substation.

Attachment 2 shows a one-line diagram of the connection facilities that exist for the (AD1-113) generation project. As indicated, it was studied as a 820 MW injection to the Jersey Central 230 kV system at the existing 230 kV connection for the Woodbridge generation plant at JCP&L’s Raritan River substation. The JCP&L facilities required to be upgraded for the primary direct connection, non-direct connection, and network upgrades of the generation project and the associated cost estimate are outlined below.

## Cost Summary

The AD1-113 “Raritan River 230 kV” project will be responsible for the following costs:

| Description                                    | Total Cost        |
|------------------------------------------------|-------------------|
| Attachment Facilities                          | \$ 0              |
| Direct Connection Network Upgrades             | \$ 0              |
| Non Direct Connection Network Upgrades         | \$ 168,800        |
| New System Upgrades                            | \$ 0              |
| Contribution to Previously Identified Upgrades | \$ 0              |
| <b>Total Costs</b>                             | <b>\$ 168,800</b> |

The transmission and substation costs given above exclude any applicable state or federal taxes. If at a future date Federal CIAC (contribution in aid of construction) taxes are deemed necessary by the IRS for this project, JCP&L shall be reimbursed by the Interconnection Customer for such taxes. JPCL estimates the tax, if applicable, would be approximately **\$26,000**.

The required Attachment Facilities, Direct Connection, and Non-Direct Connection work for the interconnection of the AD1-113 generation uprate project to the JCP&L Transmission System is detailed in the following sections. The associated one-line with the generation project is shown in Attachment 2.

Note that the FE findings were made from a conceptual review of this project. A more detailed review of the connection facilities and their cost will be identified in the System Impact and Facilities Studies. Further note that the cost estimate data contained in this document should be considered high level estimates since it was produced without a detailed engineering review. The applicant will be responsible for the actual cost of construction. FE herein reserves the right to return to any issues in this document and, upon appropriate justification, request additional monies to complete any reinforcements to the transmission systems.

## Attachment Facilities

There is no Attachment Facilities work required for this project.

## Direct Connection Cost Estimate

There is no Direct Connection scope of work required for this project.

## Non-Direct Connection Cost Estimate

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

| Description                                                                                                                        | Total Cost |
|------------------------------------------------------------------------------------------------------------------------------------|------------|
| Make relay setting changes on the 230kV line terminals. @ Raritan River SS<br><i>(PJM Network Upgrade Number n5752)</i>            | \$ 42,200  |
| Make relay setting changes on the 230kV line terminals to Raritan River. @ Parlin SS<br><i>(PJM Network Upgrade Number n5753)</i>  | \$ 42,200  |
| Make relay setting changes on the 230kV line terminals to Raritan River. @ Red Oak SS<br><i>(PJM Network Upgrade Number n5754)</i> | \$ 42,200  |

| Description                                                                                                                    | Total Cost        |
|--------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Make relay setting changes on the 230kV line terminals to Raritan River. @ Werner SS<br><br>(PJM Network Upgrade Number n5755) | \$ 42,200         |
| <b>Total Non-Direct Connection Facility Costs</b>                                                                              | <b>\$ 168,800</b> |

The total Non-Direct Connection cost estimate for the AD1-113 project is approximately **\$168,800**. This work is to perform required relay setting changes at JCP&L's Raritan River, Parlin, Red Oak, and Werner Substations.

## Schedule

Based on the extent of the JCP&L primary Non-Direct Connection work required to support the AD1-113 generation project, it is expected to take a **minimum of 7 months** from the date of a fully executed Interconnection Construction Service Agreement to complete the installation subject to market conditions and vendor lead times. This includes the requirement for the Interconnection Customer to make a preliminary payment to JCP&L which funds the construction of the Non-Direct Connection facilities work. It assumes that there will be no environmental or permitting issues to implement the upgrades required for this uprate project and that all system outages will be allowed when requested.

## JCP&L Analysis

### Power Flow Analysis

PJM performed a power flow analysis of the transmission system using a 2021 summer peak load flow model and a 2021 winter peak load flow model and the results were verified by FE. Additionally, FE performed an analysis of its underlying transmission <100 kV system. At the Primary POI, the AD1-113 project does not contribute to overloads on the FE transmission system.

### Power Factor Requirements

The existing 785 MW portion of the Customer Facility shall retain its existing ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.90 lagging (supplying VARs) measured at the generator's terminals. The increase of 35 MW to the Customer Facilities associated with the AD1-113 project shall be designed with the ability to maintain a Power Factor of at least 1.0 (unity) to 0.90 lagging (supplying VARs) measured at the generator's terminals.

FE performed a reactive capability analysis of the proposed incremental increase. The following table(s) show the reactive power requirements and capabilities for both the existing and additional output.

|                                         | Summer Leading | Winter Leading | Summer Lagging | Winter Lagging |
|-----------------------------------------|----------------|----------------|----------------|----------------|
| Generator Old Gross MW                  | 745.0          | 805.0          | 745.0          | 805.0          |
| MVar Requirement at Old Gross           | 244.9          | 264.6          | 360.8          | 389.9          |
| Reactive Power Requirement for increase | 0.0            | 0.0            | 36.3           | 17.0           |
| Total Reactive Requirement              | 244.9          | 264.6          | 397.1          | 406.8          |
| Generator New Gross                     | 820.0          | 840.0          | 820.0          | 840.0          |
| MVar Capability at New Gross            | 390.8          | 337.4          | 390.8          | 337.4          |
| Total Reactive Deficiency               | 0.0            | 0.0            | 6.3            | 69.4           |

The customer has indicated that they will cure this reactive deficiency on their side of the Point of Interconnection.

Note that a further conclusion of this study is that it will be mandatory for the Woodbridge generation plant to retain the power factor requirements stated in the Interconnection Service Agreements for the existing generation. Should AD1-113 fail to provide dynamic reactive capability from the generation project for any reason once interconnected, the FE and/or PJM Dispatchers may need to take action to curtail its output to prevent non-compliance with the power factor criteria. Schedule F of the ISA outlines the power factor requirements for AD1-113, the required testing procedures, and the consequences of not meeting the power factor requirements identified.

### **Short Circuit and Dynamics Analysis**

In accordance with the RTEP process, short circuit and dynamics analysis will not be conducted by PJM since the (AD1-113) generation project capacity increase request has no change to the electrical characteristics of the CPV Shore, LLC generating facility as indicated by the developer.

### **Stability Analysis**

A dynamic stability analysis was completed by PJM and the results were reviewed by FE. The results of the stability will be provided to the IC in PJM's System Impact Study report. There were no stability concerns identified for the faults studied.

### **System Protection Analysis**

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

## **Compliance Issues**

The proposed Customer Facilities must be designed in accordance with FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated 230 kV circuit breaker to protect the AD1-113 generator lead line. A single circuit breaker must be used to protect this line; if the project has several GSU transformers, the individual GSU transformer breakers cannot be used to protect this line.
2. The purchase and installation of the minimum required FE generation interconnection relaying and control facilities. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition ("SCADA") equipment to provide information in a compatible format to the FE Transmission System Control Center.
4. Compliance with the FE and PJM generator power factor and voltage control requirements.
5. The execution of a back-up service agreement to serve the customer load supplied from the AD1-113 generation project metering point when the units are out-of-service. This assumes the intent of the IC is to net the generation with the load.

The IC will also be required to meet all PJM, ReliabilityFirst, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and ReliabilityFirst audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the FE system.

## **CPV Shore, LLC Requirements**

CPV Shore, LLC will be responsible for meeting all criteria as specified in the applicable sections of the FE "Requirements for Transmission Connected Facilities" document including compliance with the FE and PJM generator power factor and voltage control requirements.

<http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>

The above requirements are in addition to any metering or other requirements imposed by PJM.

## **Revenue Metering and SCADA Requirements**

### **PJM Requirements**

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

### **JCP&L Requirements**

The IC will be required to comply with all FE revenue metering requirements for generation interconnection customers which can be found in FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>.

## Network Impacts

The Queue Project AD1-113 was evaluated as a 35 MW MFO (Capacity 75 MW) uprate injection into the Raritan River 230 kV bus. Project AD1-113 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AD1-113 was studied with a commercial probability of 53%. Potential network impacts were as follows:

### Summer Peak Analysis - 2021

**Generator Deliverability**

*(Single or N-1 contingencies for the Capacity portion only of the interconnection)*

None

**Multiple Facility Contingency**

*(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)*

None

**Contribution to Previously Identified Overloads**

*(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)*

None

**Steady-State Voltage Requirements**

*(Results of the steady-state voltage studies should be inserted here)*

None

**Stability and Reactive Power Requirement for Low Voltage Ride Through**

*(Summary of the VAR requirements based upon the results of the dynamic studies)*

- Reactive Power Requirement: A reactive deficiency has been identified for the 0.9 lag requirement as shown below. The customer has indicated that they will cure this reactive deficiency on their side of the POI. Schedule F of the ISA will address the need to cure this deficiency and the consequences if it is not cured.

|                                         | Summer<br>Leading | Winter<br>Leading | Summer<br>Lagging | Winter<br>Lagging |
|-----------------------------------------|-------------------|-------------------|-------------------|-------------------|
| Generator Old Gross MW                  | 745.0             | 805.0             | 745.0             | 805.0             |
| MVar Requirement at Old Gross           | 244.9             | 264.6             | 360.8             | 389.9             |
| Reactive Power Requirement for increase | 0.0               | 0.0               | 36.3              | 17.0              |
| Total Reactive Requirement              | 244.9             | 264.6             | 397.1             | 406.8             |
| Generator New Gross                     | 820.0             | 840.0             | 820.0             | 840.0             |
| MVar Capability at New Gross            | 390.8             | 337.4             | 390.8             | 337.4             |
| Total Reactive Deficiency               | 0.0               | 0.0               | 6.3               | 69.4              |

- Stability Analysis: No mitigations were found to be required for Stability. (See Attachment 3 for Stability Study Report.)

**Short Circuit**

*(Summary of impacted circuit breakers)*

None

**Affected System Analysis & Mitigation**

**NYISO Impacts:**

NYISO Impacts to be determined during later study phases (as applicable).

**Delivery of Energy Portion of Interconnection Request**

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

None

**Light Load Analysis - 2021**

No light load thermal impacts.

**Winter Analysis – 2021**

No winter thermal impacts.

**System Reinforcements**

**Short Circuit**

*(Summary form of Cost allocation for breakers will be inserted here if any)*

None

**Stability and Reactive Power Requirement**

*(Results of the dynamic studies should be inserted here)*

None

**Summer Peak Load Flow Analysis Reinforcements**

**New System Reinforcements**

*(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)*

None

**Contribution to Previously Identified System Reinforcements**

*(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)*

*(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)*

None

**Light Load Load Flow Analysis Reinforcements**

**New System Reinforcements**

*(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)*

None

**Contribution to Previously Identified System Reinforcements**

*(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)*  
*(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)*

None

**Winter Load Flow Analysis Reinforcements**

**New System Reinforcements**

*(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)*

None

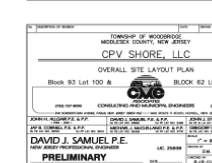
**Contribution to Previously Identified System Reinforcements**

*(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)*  
*(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)*

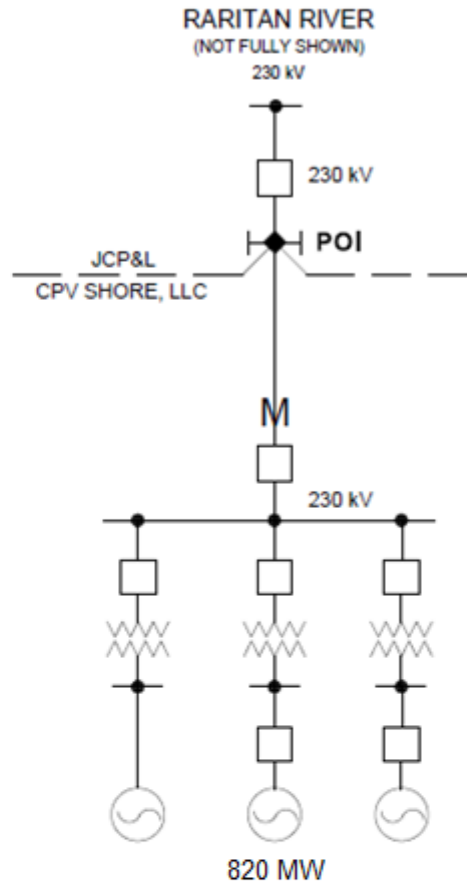
None

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## Project Location



## Attachment 2: (AD1-113) Generation Project Planning Interconnection Single Line Diagram



◆ = POI (POINT OF INTERCONNECTION) LOCATED AT JERSEY CENTRAL POWER & LIGHT (JCP&L) SUBSTATION DEAD-END STRUCTURE, WHERE CPV SHORE, LLC TRANSMISSION LINE TERMINATES.

M = REVENUE METERING FOR CPV SHORE, LLC IS OWNED, OPERATED, AND MAINTAINED BY CPV SHORE, LLC.

|                                                          |                    |                                                                                                                                          |                                 |
|----------------------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>FirstEnergy</b><br>Energy Delivery Technical Services |                    | TITLE<br>JERSEY CENTRAL POWER & LIGHT (JCP&L) RARITAN SUBSTATION 230 kV<br>INTERCONNECTION TO CPV SHORE, LLC OWNED GENERATION FACILITIES |                                 |
| BY<br>J.L.R.                                             | DATE<br>07/26/2018 | APPROVED<br>-                                                                                                                            | POI-JC-AD1-113.dwg<br>REV.<br>- |

**Attachment 3: (AD1-113) Generation Project**  
**Dynamic Stability Analysis Results**

**AD1-113**  
**System Impact Study**

**Dynamic Simulation Analysis**

Version 0

April 6, 2018

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## Executive Summary

Generator Interconnection Request AD1-113 project is for a 35 MW Maximum Facility Output (MFO) uprate to the W4-009 interconnection request. After the uprate, AD1-113 is for a 820MW MFO 2 Combustine Turbines and 1 Steam Turbines combine cycle generating facility with the Point of Interconnection (POI) on Raritan River 230kV station in JCPL system located in Middlesex County, New Jersey.

This report describes a dynamic simulation analysis of AD1-113 as part of the overall system impact study.

The load flow scenario for the analysis was based on the RTEP 2021 Light Load case, modified to include applicable queue projects. AD1-113 has been dispatched online at maximum power output, with regulating Raritan River 230kV bus voltage at 1.008 p.u., which is consistent with the default generator reference voltage specified in PJM Manual 03 Transmission Operations Section 3.3.3 for generator connections to the PJM 230 kV system.

The AD1-113 queue project was tested for compliance with NERC, PJM and other applicable criteria. The range of contingencies evaluated was limited to that necessary to assess compliance and each was limited to a 20-second simulation time period.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are not applicable.

Other applicable criteria tested include:

1. Transmission Owner (TO) specific criteria
2. Other criteria

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The system was tested for a system intact condition and the fault types listed above. Specific fault descriptions and breaker clearing times used for this study are provided in the result table.

For all simulations, the queue project under study along with the rest of the PJM system were required to maintain synchronism and with all states returning to an acceptable new condition following the disturbance.

For the fault contingencies tested on the 2021 light load case:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 3%.
- b) The AD1-113 generator was able to ride through all faults (except for faults where protective action trips a generator(s)).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

No mitigations were found to be required.

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## **1. Introduction**

Generator Interconnection Request AD1-113 project is for a 35 MW Maximum Facility Output (MFO) uprate to the W4-009 interconnection request. After the uprate, AD1-113 is for a 820MW MFO 2 Combustion Turbines and 1 Steam Turbines combine cycle generating facility with the Point of Interconnection (POI) on Raritan River 230kV station in JCPL system located in Middlesex County, New Jersey.

This analysis is effectively a screening study to determine whether the addition of AD1-113 will meet the dynamic requirements of the NERC, PJM and Transmission Owner reliability standards.

In this report the AD1-113 project and how it is proposed to be connected to the grid are first described, followed by a description of how the project is modeled in this study. The fault cases are then described and analyzed, and lastly a discussion of the results is provided.

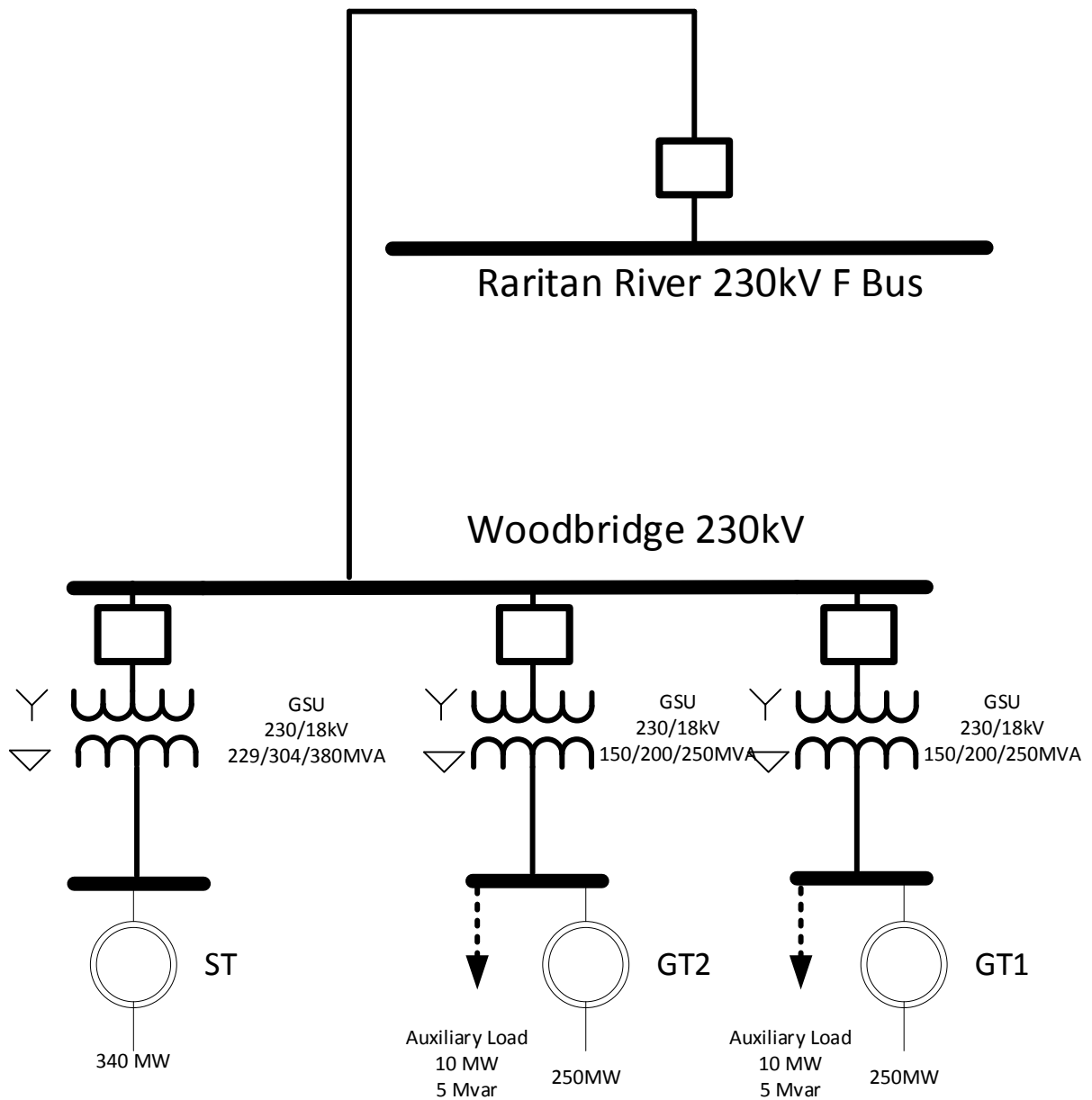
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## **2. Description of Project**

Generator Interconnection Request AD1-113 project is for a 35 MW Maximum Facility Output (MFO) uprate to the W4-009 interconnection request. After the uprate, AD1-113 is for a 820MW MFO 2 Combustine Turbines and 1 Steam Turbines combine cycle generating facility with the Point of Interconnection (POI) on Raritan River 230kV station in JCPL system located in Middlesex County, New Jersey.

Figure 1 shows the simplified one-line diagram of the AD1-113 loadflow model. Table 1 lists the parameters given in the impact study data and the corresponding parameters of the AD1-113 loadflow model.

The dynamic models for AD1-113 plant are based on the PSS\E standard model with parameters supplied by the developer.



**Figure 1: AD1-113 Plant Model**

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**Table 1: AD1-113 Plant Model**

|            | Impact Study Data                                         | Model                      |
|------------|-----------------------------------------------------------|----------------------------|
| Generators | 2 X 250MW GT                                              | 2 × 250 MW GE GT           |
|            | MVA base = 262 MVA                                        | Pgen 250 MW                |
|            | Vt = 18 kV                                                | Pmax 250 MW                |
|            |                                                           | Pmin 0 MW                  |
|            | Unsaturated sub-transient reactance = 0.205 pu @ MVA base | Qmax 135 MVA <sub>r</sub>  |
|            |                                                           | Qmin -90 MVA <sub>r</sub>  |
|            |                                                           | Mbase 262 MVA              |
|            |                                                           | Zsorce j0.205 pu @ Mbase   |
|            |                                                           |                            |
|            | 1 × 340 MW ST                                             | 1 × 340 MW ST              |
|            | MVA base = 385 MVA                                        | Pgen 340 MW                |
|            | Vt = 18 kV                                                | Pmax 340 MW                |
|            |                                                           | Pmin 0 MW                  |
|            | Unsaturated sub-transient reactance = j0.25 pu @ MVA base | Qmax 165 MVA <sub>r</sub>  |
|            |                                                           | Qmin -135 MVA <sub>r</sub> |
|            |                                                           | Mbase 385 MVA              |
|            |                                                           | Zsorce j0.25 pu @ Mbase    |

|                    | Impact Study Data                          | Model                                      |
|--------------------|--------------------------------------------|--------------------------------------------|
| GSU transformer(s) | GT1 GSU 230/18 kV two winding transformer  | GT1 GSU 230/18 kV two winding transformer  |
|                    | Rating = 150/200/250 MVA                   | Rating = 150/200/250 MVA                   |
|                    | Transformer base = 150 MVA                 | Transformer base = 150 MVA                 |
|                    | Impedance = 0.00142 + j0.0934pu @ MVA base | Impedance = 0.00142 + j0.0934pu @ MVA base |
|                    | Number of taps = N/A                       | Number of taps = 5                         |
|                    | Tap step size = N/A                        | Tap step size = 2.5%                       |
|                    | GT2 GSU 230/18 kV two winding transformer  | GT2 GSU 230/18 kV two winding transformer  |
|                    | Rating = 150/200/250 MVA                   | Rating = 150/200/250 MVA                   |
|                    | Transformer base = 150 MVA                 | Transformer base = 150 MVA                 |
|                    | Impedance = 0.00142 + j0.0934pu @ MVA base | Impedance = 0.00142 + j0.0934pu @ MVA base |
|                    | Number of taps = N/A                       | Number of taps = 5                         |
|                    | Tap step size = N/A                        | Tap step size = 2.5%                       |
|                    | Page 9                                     |                                            |
|                    | ST GSU 230/18kV tho winding transformer    | ST GSU 230/18kV tho winding transformer    |

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|                 | <b>Impact Study Data</b>               | <b>Model</b>                                            |
|-----------------|----------------------------------------|---------------------------------------------------------|
| Auxiliary load  | 2 x 10 MW, 5 MVA <sub>r</sub>          | 2 x 10 MW, 5 MVA <sub>r</sub>                           |
| Station load    | N/A                                    | N/A                                                     |
| Attachment line | Not Provided, use as built in the case | R = 0.00208 p.u.<br>X = 0.02288 p.u.<br>B = 0.0074 p.u. |

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### 3. Loadflow and Dynamics Case Setup

The dynamics simulation analysis was carried out using PSS/E Version 33.7.

The load flow scenario and fault cases for this study are based on PJM's Regional Transmission Planning Process<sup>1</sup>.

The selected load flow scenario is the RTEP 2021 light load case with the following modifications:

- a) Addition of all applicable queue projects prior to AD1-113.
- b) Addition of AD1-113 queue project.
- c) Removal of withdrawn and subsequent queue projects in the vicinity of AD1-113.
- d) Dispatch of units in the PJM system to maintain slack generators within limits.

The AD1-113 initial conditions are listed in Table 2, indicating maximum power output with regulating Raritan River 230kV bus voltage at 1.008 p.u., which is consistent with the default generator reference voltage specified in PJM Manual 03 Transmission Operations Section 3.3.3 for generator connections to the PJM 230 kV system.

**Table 2: AD1-113 machine initial conditions**

| <b>Bus</b> | <b>Name</b> | <b>Unit</b> | <b>PGEN</b> | <b>QGEN</b> | <b>ETERM</b> | <b>POI Voltage</b> |
|------------|-------------|-------------|-------------|-------------|--------------|--------------------|
| 934841     | AD1-113 CT1 | 1           | 250         | 17.4        | 0.99         | 1.008              |
| 934842     | AD1-113 CT2 | 1           | 250         | 17.4        | 0.99         | 1.008              |
| 934843     | AD1-113 ST  | 1           | 250         | 17.4        | 0.99         | 1.008              |

Generation within the PJM500 system (area 225 in the PSS/E case) and within the vicinity of AD1-113 has been dispatched online at maximum output (P<sub>MAX</sub>). The dispatch of generation in the vicinity of AD1-113 is given in Attachment 3.

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<sup>1</sup> Manual 14B: PJM Region Transmission Planning Process, Rev 33, May 5 2016, Attachment G : PJM Stability, Short Circuit, and Special RTEP Practices and Procedures.

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## 4. Fault Cases

Tables 3 listed the contingencies and results that were studied, with representative worst case total clearing times provided by PJM. Each contingency was studied over a 20 second simulation time interval.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are not applicable.

Other applicable criteria tested include:

1. Transmission Owner (TO) specific criteria
2. Other criteria

The system was tested for a system intact condition and the fault types listed above.

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## 5. Evaluation Criteria

This study is focused on AD1-113, along with the rest of the PJM system, maintaining synchronism and having all states return to an acceptable new condition following the disturbance. The recovery criteria applicable to this study are as per PJM's Regional Transmission Planning Process and Transmission Owner criteria:

- a) The system with AD1-113 included is transiently stable and post-contingency oscillations should be positively damped with a damping margin of at least 3%
- b) The AD1-113 is able to ride through faults (except for faults where protective action trips AD1-113).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

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## 6. Summary of Results

Plots from the dynamic simulations are provided in Attachment 4, with results summarized in Table 3.

For the studied fault contingencies tested on the 2021 light load case:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 3%.
- b) The AD1-113 generator was able to ride through all faults (except for faults where protective action trips a generator(s)).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

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## **7. System Reinforcement Requirements**

None.

**Table 3: Contingency List and Stability Study Results****P0: Stead State**

| <b>Fault ID</b> | <b>Duration</b>         |
|-----------------|-------------------------|
| P0.00           | Steady State 20 sec run |

**P1: Three Phase Faults with normal clearing**

| <b>Fault ID</b> | <b>Fault description</b>                                                                        | <b>Clearing Time Normal &amp; HSR (Cycles)</b> | <b>Results</b> |
|-----------------|-------------------------------------------------------------------------------------------------|------------------------------------------------|----------------|
| P1.00           | 3ph @ Raritan River – Woodbridge 230kV, normal clear loss of AD1-113                            | 5                                              | Stable         |
| P1.01           | 3ph @ Raritan River Sayerville GSU, normal clear loss of Raritan CT 1-4                         | 5                                              | Stable         |
| P1.02           | 3ph @ Raritan River – Red Oak – South River Junction, normal clear loss of Red Oak unit 2 and 3 | 5                                              | Stable         |
| P1.03           | 3ph @ Raritan River – Red Oak – South River Junction, normal clear loss of Red Oak unit 1 and 4 | 5                                              | Stable         |
| P1.04           | 3ph @ Raritan River – NJ Steel – Werner 230kV, normal clear                                     | 5                                              | Stable         |
| P1.05           | 3ph @ Raritan River – Kilmer – Lake Nelson 230kV ckt 1, normal clear                            | 5                                              | Stable         |
| P1.06           | 3ph @ Raritan River – Kilmer – Lake Nelson 230kV ckt 2, normal clear                            | 5                                              | Stable         |
| P1.07           | 3ph @ Raritan River 230kV, normal clear loss of Raritan River 230/115kV TF 17                   | 5                                              | Stable         |
| P1.08           | 3ph @ Raritan River 230kV, normal clear loss of Raritan River 230/115kV TF 13                   | 5                                              | Stable         |
| P1.09           | 3ph @ Raritan River – Neptune Converter 230kV, normal clear                                     | 5                                              | Stable         |
| P1.10           | 3ph @ Raritan River 115kV, normal clear loss of Raritan River 230/115kV TF 17                   | 7                                              | Stable         |
| P1.11           | 3ph @ Raritan River 115kV, normal clear loss of Raritan River 230/115kV TF 13                   | 7                                              | Stable         |
| P1.12           | 3ph @ Raritan River 115kV, normal clear loss of Raritan River 115/34.5kV TF 4                   | 7                                              | Stable         |
| P1.13           | 3ph @ Raritan River 115kV, normal clear loss of Raritan River 115/34.5kV TF 14                  | 7                                              | Stable         |
| P1.14           | 3ph @ Raritan River – Werner 115kV, normal clear                                                | 7                                              | Stable         |

| <b>Fault ID</b> | <b>Fault description</b>                                                                              | <b>Clearing Time Normal &amp; HSR (Cycles)</b> | <b>Results</b> |
|-----------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------|----------------|
| P1.15           | 3ph @ Raritan River – Deep Run 115kV ckt 1, normal clear                                              | 7                                              | Stable         |
| P1.16           | 3ph @ Raritan River – Deep Run 115kV ckt 2, normal clear                                              | 7                                              | Stable         |
| P1.17           | 3ph @ Lake Nelson – Kilmer – Raritan River 230kV ckt 1, normal clear                                  | 5                                              | Stable         |
| P1.18           | 3ph @ Lake Nelson – Kilmer – Raritan River 230kV ckt 2, normal clear                                  | 5                                              | Stable         |
| P1.19           | 3ph @ Lake Nelson – Green Brook 230kV, normal clear                                                   | 5                                              | Stable         |
| P1.20           | 3ph @ Lake Nelson – Middlesex 230kV, normal clear                                                     | 5                                              | Stable         |
| P1.21           | 3ph @ Green Brook – Lake Nelson 230kV, normal clear                                                   | 5                                              | Stable         |
| P1.22           | 3ph @ Green Brook – Gillette 230kV, normal clear                                                      | 5                                              | Stable         |
| P1.23           | 3ph @ Middlesex – Lake Nelson 230kV, normal clear                                                     | 5                                              | Stable         |
| P1.24           | 3ph @ Middlesex – Bridgewater 230kV, normal clear                                                     | 5                                              | Stable         |
| P1.25           | 3ph @ Middlesex – Green Brook – Gillette 230kV, normal clear                                          | 5                                              | Stable         |
| P1.26           | 3ph @ South River Junction – Red Oak – Raritan River 230kV, normal clear loss of Red Oak unit 2 and 3 | 5                                              | Stable         |
| P1.27           | 3ph @ South River Junction – Red Oak – Raritan River 230kV, normal clear loss of Red Oak unit 1 and 4 | 5                                              | Stable         |
| P1.28           | 3ph @ South River Junction – AC1-029 230kV, normal clear                                              | 5                                              | Stable         |
| P1.29           | 3ph @ South River Junction – South River 230kV, normal clear                                          | 5                                              | Stable         |
| P1.30           | 3ph @ South River Junction – Parlin 230kV, normal clear                                               | 5                                              | Stable         |
| P1.31           | 3ph @ South River Junction – Atlantic 230kV, normal clear                                             | 5                                              | Stable         |
| P1.32           | 3ph @ AC1-029 POI – South River 230kV, normal clear                                                   | 5                                              | Stable         |
| P1.33           | 3ph @ AC1-029 POI – AC1-029 Main 230kV, normal clear loss of AC1-029                                  | 5                                              | Stable         |
| P1.34           | 3ph @ AC1-029 POI – South River Junction 230kV, normal clear                                          | 5                                              | Stable         |
| P1.35           | 3ph @ South River – South River Junction 230kV, normal clear                                          | 5                                              | Stable         |
| P1.36           | 3ph @ South River – AC1-029 POI 230kV, normal clear                                                   | 5                                              | Stable         |
| P1.37           | 3ph @ South River 230kV, normal clear loss of South River Unit 1                                      | 5                                              | Stable         |
| P1.38           | 3ph @ South River 230kV, normal clear loss of South River Unit 2 and 3                                | 5                                              | Stable         |
| P1.39           | 3ph @ Parlin – South River Junction 230kV, normal clear                                               | 5                                              | Stable         |

| <b>Fault ID</b> | <b>Fault description</b>                                                                 | <b>Clearing Time Normal &amp; HSR (Cycles)</b> | <b>Results</b> |
|-----------------|------------------------------------------------------------------------------------------|------------------------------------------------|----------------|
| P1.40           | 3ph @ Parlin – Williams Gas – Freneau 230kV, normal clear                                | 5                                              | Stable         |
| P1.41           | 3ph @ Parlin 230kV, normal clear loss of Parlin unit 1 and 2, Parlin 230kV load (Dupont) | 5                                              | Stable         |
| P1.42           | 3ph @ Parlin 230kV, normal clear loss of Parlin unit 3 and 4                             | 5                                              | Stable         |

#### **P4: SLG Stuck Breaker (SB) Faults at Backup Clearing**

| <b>Fault ID</b> | <b>Fault description</b>                                                                                                                                                                                              | <b>Clearing Time Near &amp; Remote (Cycles)</b> | <b>Results</b> |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------|
| P4.01           | SLG @ Raritan River 230kV, normal clear loss of Sayerville units, Raritan River 230/115kV TF 17, SB @ Raritan River, delayed clear loss of Raritan River – Neptune Converter 230kV                                    | 5 / 16                                          | Stable         |
| P4.02           | SLG @ Raritan River 230kV, normal clear loss of Sayerville units, Raritan River 230/115kV TF 17, SB @ Raritan River, delayed clear loss of Raritan River – Kilmer – Lake Nelson 230kV ckt 1                           | 5 / 16                                          | Stable         |
| P4.03           | SLG @ Raritan River 230kV, normal clear loss of Sayerville units, Raritan River 230/115kV TF 17, SB @ Raritan River, delayed clear loss of Raritan River – Kilmer – Lake Nelson 230kV ckt 2                           | 5 / 16                                          | Stable         |
| P4.04           | SLG @ Raritan River 230kV, normal clear loss of Sayerville units, Raritan River 230/115kV TF 17, SB @ Raritan River, delayed clear loss of Raritan River – NJ Steel – Werner 230kV                                    | 5 / 16                                          | Stable         |
| P4.05           | SLG @ Raritan River 230kV, normal clear loss of Sayerville units, Raritan River 230/115kV TF 17, SB @ Raritan River, delayed clear loss of Raritan River – Red Oak – South River Junction 230kV, Red Oak unit 1 and 4 | 5 / 16                                          | Stable         |
| P4.06           | SLG @ Raritan River 230kV, normal clear loss of Sayerville units, Raritan River 230/115kV TF 17, SB @ Raritan River, delayed clear loss of Raritan River – Red Oak – South River Junction 230kV, Red Oak unit 2 and 3 | 5 / 16                                          | Stable         |
| P4.07           | SLG @ Raritan River 230kV, normal clear loss of Raritan River – Woodbridge 230kV, AD1-113, Raritan River 230/115kV TF 13, SB @ Raritan River, delayed clear loss of Raritan River – Neptune Converter 230kV           | 5 / 16                                          | Stable         |
| P4.08           | SLG @ Raritan River 230kV, normal clear loss of Raritan River – Woodbridge 230kV, AD1-113, Raritan River 230/115kV TF 13, SB @ Raritan River, delayed clear loss of Raritan River – Kilmer – Lake Nelson 230kV ckt 1  | 5 / 16                                          | Stable         |

| <b>Fault ID</b> | <b>Fault description</b>                                                                                                                                                                                                                       | <b>Clearing Time Near &amp; Remote (Cycles)</b> | <b>Results</b> |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------|
| P4.09           | SLG @ Raritan River 230kV, normal clear loss of Raritan River – Woodbridge 230kV, AD1-113, Raritan River 230/115kV TF 13, SB @ Raritan River, delayed clear loss of Raritan River – Kilmer – Lake Nelson 230kV ckt 2                           | 5 / 16                                          | Stable         |
| P4.10           | SLG @ Raritan River 230kV, normal clear loss of Raritan River – Woodbridge 230kV, AD1-113, Raritan River 230/115kV TF 13, SB @ Raritan River, delayed clear loss of Raritan River – NJ Steel – Werner 230kV                                    | 5 / 16                                          | Stable         |
| P4.11           | SLG @ Raritan River 230kV, normal clear loss of Raritan River – Woodbridge 230kV, AD1-113, Raritan River 230/115kV TF 13, SB @ Raritan River, delayed clear loss of Raritan River – Red Oak – South River Junction 230kV, Red Oak unit 1 and 4 | 5 / 16                                          | Stable         |
| P4.12           | SLG @ Raritan River 230kV, normal clear loss of Raritan River – Woodbridge 230kV, AD1-113, Raritan River 230/115kV TF 13, SB @ Raritan River, delayed clear loss of Raritan River – Red Oak – South River Junction 230kV, Red Oak unit 2 and 3 | 5 / 16                                          | Stable         |
| P4.13           | SLG @ Raritan River 230kV, normal clear loss of Raritan River 230/115kV TF 17, SB @ Raritan River, delayed clear loss of Sayerville units                                                                                                      | 5 / 16                                          | Stable         |
| P4.14           | SLG @ Raritan River 230kV, normal clear loss of Sayerville units, SB @ Raritan River, delayed clear loss of Raritan River 230/115kV TF 17                                                                                                      | 5 / 16                                          | Stable         |
| P4.15           | SLG @ Raritan River 230kV, normal clear loss of Raritan 230/115kV TF 13, SB @ Raritan River, delayed clear loss of Raritan River – Woodbridge 230kV, AD1-113                                                                                   | 5 / 16                                          | Stable         |
| P4.16           | SLG @ Raritan River 230kV, normal clear loss of Raritan River – Woodbridge 230kV, AD1-113, SB @ Raritan River, delayed clear loss of Raritan 230/115kV TF 13                                                                                   | 5 / 16                                          | Stable         |
| P4.17           | SLG @ Raritan River 115kV, normal clear loss of Raritan River 115/34.5kV TF14, SB @ Raritan River 115kV, delayed clear loss of Raritan River 230/115kV TF 17                                                                                   | 7 / 20                                          | Stable         |
| P4.18           | SLG @ Raritan River 115kV, normal clear loss of Raritan River 115/34.5kV TF14, SB @ Raritan River 115kV, delayed clear loss of Raritan River – Deep Run 115kV ckt 1                                                                            | 7 / 20                                          | Stable         |
| P4.19           | SLG @ Raritan River 115kV, normal clear loss of Raritan River 115/34.5kV TF4, SB @ Raritan River 115kV, delayed clear loss of Raritan River 230/115kV TF 17                                                                                    | 7 / 20                                          | Stable         |
| P4.20           | SLG @ Raritan River 115kV, normal clear loss of Raritan River 115/34.5kV TF4, SB @ Raritan River 115kV, delayed clear loss of Raritan River 230/115kV TF 13                                                                                    | 7 / 20                                          | Stable         |
| P4.21           | SLG @ Raritan River 115kV, normal clear loss of Raritan River – Deep Run ckt 2, SB @ Raritan River 115kV, delayed clear loss of Raritan River – Deep Run 115kV ckt 1                                                                           | 7 / 20                                          | Stable         |

| <b>Fault ID</b> | <b>Fault description</b>                                                                                                                                                                                                                                                 | <b>Clearing Time Near &amp; Remote (Cycles)</b> | <b>Results</b> |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------|
| P4.22           | SLG @ Raritan River 115kV, normal clear loss of Raritan River – Deep Run ckt 2, SB @ Raritan River 115kV, delayed clear loss of Raritan River – Werner 115kV                                                                                                             | 7 / 20                                          | Stable         |
| P4.23           | SLG @ Lake Nelson 230kV, normal clear loss of Lake Nelson – Green Brook 230kV, SB @ Lake Nelson, delayed clear loss of Lake Nelson – Kilmer – Raritan River 230kV ckt 2                                                                                                  | 5 / 17                                          | Stable         |
| P4.24           | SLG @ Lake Nelson 230kV, normal clear loss of Lake Nelson – Middlesex 230kV, SB @ Lake Nelson, delayed clear loss of Lake Nelson – Kilmer – Raritan River 230kV ckt 1                                                                                                    | 5 / 17                                          | Stable         |
| P4.25           | SLG @ South River Junction 230kV, normal clear loss of South River Junction – Red Oak – Raritan River 230kV line, Red Oak unit 2 and 3, SB @ South River Junction, delayed clear loss of South River Junction – Red Oak – Raritan River 230kV line, Red Oak unit 1 and 4 | 5 / 16                                          | Stable         |
| P4.26           | SLG @ South River Junction 230kV, normal clear loss of South River Junction – Red Oak – Raritan River 230kV line, Red Oak unit 2 and 3, SB @ South River Junction, delayed clear loss of South River Junction – Parlin 230kV                                             | 5 / 16                                          | Stable         |
| P4.27           | SLG @ South River Junction 230kV, normal clear loss of South River Junction – Atlantic 230kV, SB @ South River Junction, delayed clear loss of South River Junction – Parlin 230kV                                                                                       | 5 / 16                                          | Stable         |
| P4.28           | SLG @ South River Junction 230kV, normal clear loss of South River Junction – Atlantic 230kV, SB @ South River Junction, delayed clear loss of South River Junction – South River 230kV                                                                                  | 5 / 16                                          | Stable         |
| P4.29           | SLG @ South River Junction 230kV, normal clear loss of South River Junction – AC1-029 POI 230kV, SB @ South River Junction, delayed clear loss of South River Junction – South River 230kV, loss of AC1-029 and South River generators                                   | 5 / 16                                          | Stable         |
| P4.30           | SLG @ South River Junction 230kV, normal clear loss of South River Junction – AC1-029 POI 230kV, SB @ South River Junction, delayed clear loss of South River Junction – Red Oak – Raritan River 230kV line, Red Oak unit 1 and 4                                        | 5 / 16                                          | Stable         |
| P4.31           | SLG @ Parlin 230kV, normal clear loss of Parlin Load (Dupont), Parlin unit 1 and 2, SB @ Parlin 230kV, delayed clear loss of Parlin – South River Junction 230kV                                                                                                         | 5 / 16                                          | Stable         |
| P4.32           | SLG @ Parlin 230kV, normal clear loss of Parlin Load (Dupont), Parlin unit 1 and 2, SB @ Parlin 230kV, delayed clear loss of Parlin – Williams Gas – Freneau 230kV                                                                                                       | 5 / 16                                          | Stable         |
| P4.33           | SLG @ Parlin 230kV, normal clear loss of Parlin unit 3 and 4, SB @ Parlin 230kV, delayed clear loss of Parlin – South River Junction 230kV                                                                                                                               | 5 / 16                                          | Stable         |
| P4.34           | SLG @ Parlin 230kV, normal clear loss of Parlin unit 3 and 4, SB @ Parlin 230kV, delayed clear loss of Parlin – Williams Gas – Freneau 230kV                                                                                                                             | 5 / 16                                          | Stable         |
| P4.35           | SLG @ AC1-029 POI 230kV, normal clear loss of AC1-029 POI – South River 230 kV, SB @ AC1-029 POI, delayed clear loss of AC1-029, AC1-029 POI – South River Junction 230kV                                                                                                | 5 / 16                                          | Stable         |

| <b>Fault ID</b> | <b>Fault description</b>                                                                                                                                 | <b>Clearing Time Near &amp; Remote (Cycles)</b> | <b>Results</b> |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------|
| P4.36           | SLG @ South River 230kV, normal clear loss of South River – AC1-029 POI 230kV, SB @ South River, delayed clear loss of South River unit 1                | 5 / 16                                          | Stable         |
| P4.37           | SLG @ South River 230kV, normal clear loss of South River – AC1-029 POI 230kV, SB @ South River, delayed clear loss of South River unit 2 and 3          | 5 / 16                                          | Stable         |
| P4.38           | SLG @ South River 230kV, normal clear loss of South River – South River Junction 230kV, SB @ South River, delayed clear loss of South River unit 1       | 5 / 16                                          | Stable         |
| P4.39           | SLG @ South River 230kV, normal clear loss of South River – South River Junction 230kV, SB @ South River, delayed clear loss of South River unit 2 and 3 | 5 / 16                                          | Stable         |

#### **P5: SLG Fault with Delayed Clearing**

| <b>Fault ID</b> | <b>Fault description</b>                                                                                                               | <b>Clearing Time Near &amp; Remote (Cycles)</b> | <b>Results</b> |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------|
| P5.01           | SLG @ 80% of Raritan River – Kilmer – Lake Nelson 230kV ckt 1, delayed clear at Raritan River                                          | 5 / 34                                          | Stable         |
| P5.02           | SLG @ 80% of Lake Nelson – Kilmer – Raritan River 230kV ckt 1, delayed clear at Lake Nelson                                            | 5 / 34                                          | Stable         |
| P5.03           | SLG @ 80% of Raritan River – Kilmer – Lake Nelson 230kV ckt 2, delayed clear at Raritan River                                          | 5 / 34                                          | Stable         |
| P5.04           | SLG @ 80% of Lake Nelson – Kilmer – Raritan River 230kV ckt 2, delayed clear at Lake Nelson                                            | 5 / 34                                          | Stable         |
| P5.05           | SLG @ 80% of Raritan River – Red Oak – South River Junction 230kV, delayed clear at Raritan River, loss of Red Oak unit 2 and 3        | 5*                                              | Stable         |
| P5.06           | SLG @ 80% of South River Junction – Red Oak – Raritan River 230kV, delayed clear at South River Junction, loss of Red Oak unit 2 and 3 | 5*                                              | Stable         |
| P5.07           | SLG @ 80% of Raritan River – Red Oak – South River Junction 230kV, delayed clear at Raritan River, loss of Red Oak unit 1 and 4        | 5*                                              | Stable         |

| <b>Fault ID</b> | <b>Fault description</b>                                                                                                               | <b>Clearing Time Near &amp; Remote (Cycles)</b> | <b>Results</b> |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------|
| P5.08           | SLG @ 80% of South River Junction – Red Oak – Raritan River 230kV, delayed clear at South River Junction, loss of Red Oak unit 1 and 4 | 5*                                              | Stable         |
| P5.09           | SLG @ 80% of Raritan River – NJ Steel – Werner 230kV, delayed clear at Raritan River                                                   | 5 / 34                                          | Stable         |
| P5.10           | SLG @ 80% of Werner – NJ Steel – Raritan River 230kV, delayed clear at Werner                                                          | 5 / 34                                          | Stable         |
| P5.11           | SLG @ 80% of South River Junction – Atlantic 230kV, delayed clear at South River Junction                                              | 5 / 34                                          | Stable         |
| P5.12           | SLG @ 80% of Atlantic – South River Junction 230kV, delayed clear at Atlantic                                                          | 5 / 34                                          | Stable         |
| P5.13           | SLG @ 80% of South River Junction – Parlin 230kV, delayed clear at South River Junction                                                | 5*                                              | Stable         |
| P5.14           | SLG @ 80% of Parlin – South River Junction 230kV, delayed clear at Atlantic                                                            | 5*                                              | Stable         |
| P5.15           | SLG @ 80% of Parlin – Williams – Freneau 230kV, delayed clear at Parlin                                                                | 5*                                              | Stable         |
| P5.16           | SLG @ 80% of Freneau – Williams – Parlin 230kV, delayed clear at Parlin                                                                | 5*                                              | Stable         |

**\*Dual communication scheme exists**

## **P7: SLG Fault with Common Structure Failure**

| <b>Fault ID</b> | <b>Fault description</b>                                                                | <b>Clearing Time (Cycles)</b> | <b>Results</b> |
|-----------------|-----------------------------------------------------------------------------------------|-------------------------------|----------------|
| P7.01           | CONTINGENCY 'JC-P7-1-JCC-230-18T' /* LAKE NELSON - GREEN BROCK - GILLETTE I1023 & W1037 | 5                             | Stable         |
| P7.02           | CONTINGENCY 'JC-P7-1-JCC-230-3' /* PARLIN-RED OAK-RAR R & S.RIVER-RED OAK-RAR RIVER     | 5                             | Stable         |
| P7.03           | CONTINGENCY 'JC-P7-1-JCC-230-4' /* R11B-ATLANTIC (P1030) & PARLIN-FRENEAU (K1025)       | 5                             | Stable         |
| P7.04           | CONTINGENCY 'JC-P7-1-JCC-230-5A' /* S.RVR-ATLANTIC & FRENEAU-ATLANTIC 230 KV            | 5                             | Stable         |

| <b>Fault ID</b> | <b>Fault description</b>                                                                             | <b>Clearing Time (Cycles)</b> | <b>Results</b> |
|-----------------|------------------------------------------------------------------------------------------------------|-------------------------------|----------------|
| P7.05           | CONTINGENCY 'JC-P7-1-JCC-230-6B' /* DEEP RUN-RAR RVR B2 & C3 115 KV FUTURE DEEP RUN BREAKERS         | 5                             | Stable         |
| P7.06           | CONTINGENCY 'JC-P7-1-JCC-230-9AT' /* RAR R-NEJESCO-WERNER 230 KV & RAR R-WERNER 115 KV               | 5                             | Stable         |
| P7.07           | CONTINGENCY 'PS_P7-1_I1023+GBK-LN_LT' /* LAKE NELSON - GILLETE BRIDGEWATER & LAKE NELSON -GREENBROOK | 5                             | Stable         |
| P7.08           | CONTINGENCY 'PS_P7-1_I1023+W1037' /* LAKE NELSON - GILLETE BRIDGEWATER & GILLETE -GREENBROOK         | 5                             | Stable         |



2750 Monroe Blvd.  
Audubon, PA 19403

*Via DocuSign*

**September 7, 2018**

**CPV Shore, LLC**  
8403 Colesville Road  
Suite 915  
Silver Spring, Maryland 20910

Dear **Mr. Favinger**,

**Re: AD1-113 – “Raritan River 230 kV”- System Impact Study Report, Interconnection Service Agreement, and Interconnection Construction Service Agreement**

Enclosed is a report documenting the results of the **AD1-113** System Impact Study. The results of this study are predicated on a **2021** transmission system based upon PJM's best assumptions at the present time for load growth and connection of proposed new generation additions.

System Impact Studies are performed to determine the facilities required for interconnection and to define the estimated cost and timing for construction of attachment facilities and network upgrades required for the reliable interconnection of a generation project to the PJM system. The costs and associated timing described in the enclosed report are based upon estimates given to PJM by the affected Transmission Owner(s). The costs are your responsibility as the project developer.

The costs associated with the study are being tabulated and you will receive a final statement/invoice electronically from PJM detailing your balance within 120 days.

Please be advised that all modeling will be completed consistent with Manual 3A. Market settlements cannot begin until these steps have been complete.

Note that Tariff §212.5 milestones require that you have all site permits, water and fuel agreements and associated right of way, and a memorandum of understanding for major equipment at the time you return your executed Interconnection Service Agreement (ISA). It is your responsibility to ensure these requirements are met and if they cannot be met at the time of the return of the ISA, you must demonstrate your due diligence and propose dates when those milestones will be met.

Since it has been determined that a Facilities Study is not required for this project and pursuant to Section 212 of the PJM Tariff, an Interconnection Service Agreement (ISA), defining the project's cost responsibility for the construction of the required attachment and network facilities, specifying the project's Interconnection Rights, and obligating Interconnection Customer to abide by the rules and procedures pertaining to generation in the PJM control area, is enclosed for your execution. A Security amount of **\$168,800** is specified in the ISA. The ISA must be executed

via **DocuSign** within **60 days** of receipt of the System Impact Study Report by close of business on **November 7, 2018** in order to maintain the project's position in the queue. At the same time, the Security deposit must be received by PJM by this date. This Security deposit can be in the form of a check, wire transfer, or letter or credit.

Also included is the Interconnection Construction Service Agreement (ICSA). Pursuant to Section 212 of the PJM Tariff, the ICSA must be executed via **DocuSign** within **90 days** of receipt (by close of business **December 7, 2018**). Options to dispute or to file the ISA or ICSAs unexecuted are described in section 212 of the PJM Tariff.

Please review and execute the ISA and ICSA as specified in **DocuSign**. Once PJM receives the required security and the agreements are fully executed, PJM will process the agreements with FERC accordingly. The final executed copies of the ISA and ICSA will be available via DocuSign.

Please send the required security to:

Jeannette Mittan  
PJM Interconnection, L.L.C.  
2750 Monroe Blvd.  
Audubon, PA 19403

**The following information is provided for wire transfers:**

Bank: PNC Bank, NA, New Jersey  
ABA Number: 031-207-607  
Account Number: 8013589826

Please e-mail PJM at [SystemPlanning.Admin@pjm.com](mailto:SystemPlanning.Admin@pjm.com) with the project name, queue number, date and amount of wire.

**Letters of Credit ("L/C"):** The Letter of Credit will only be accepted from U.S.-based financial institutions or U.S. branches of foreign financial institutions that have a minimum corporate debt rating of "A" by Standard & Poor's or Fitch Ratings, or "A2" from Moody's Investors Service, or an equivalent short term rating from one of these agencies. PJM will consider the lowest applicable rating to be the rating of the financial institution. If the rating of a financial institution providing a Letter of Credit is lowered below A/A2 by any rating agency, then PJM may require the Participant to provide a Letter of Credit from another financial institution that is rated A/A2 or better, or to provide a cash deposit. If a Letter of Credit is provided from a U.S. branch of a foreign institution, the U.S. branch must itself comply with the terms of this credit policy, including having its own acceptable credit rating. Any questions regarding whether a financial institution is acceptable or not should be addressed [credit\\_hotline@pjm.com](mailto:credit_hotline@pjm.com). You may access the Standard Letter of Credit through the PJM webpage under Planning/RTEP Development/Expansion Planning Process. Any deviation from the standard Letter of Credit form must be approved by PJM before the Letter of Credit is executed. Please allow at least two weeks for Letter of Credit review if any changes from the standard form are requested. If there are any questions, please contact Jeannette Mittan (contact information in wire transfer section above).

Following the execution of this ISA, your contact at PJM will be **Chibuzor Ofoegbu** at 610.666.2375 or by e-mail at [Chibuzor.Ofoegbu@pjm.com](mailto:Chibuzor.Ofoegbu@pjm.com). Please keep him copied on correspondence with JCPL.

**Stephanie Dalton** of FirstEnergy will be your contact for the Transmission Owner organization. She can be contacted at 330.384.2477 or by e-mail at [sdalton@firstenergycorp.com](mailto:sdalton@firstenergycorp.com).

If you wish to discuss the results of the System Impact Report or the ISA/ICSA agreements with me, please let me know. My office telephone number is 610.666.4561 and my email address is [Lisa.Krizenoskas@pjm.com](mailto:Lisa.Krizenoskas@pjm.com).

Sincerely,



Lisa Krizenoskas

Sr. Engineer  
PJM Interconnection Projects

Attachments

PJM (w/attachments):

Tom Favinger-IC  
Stephanie Dalton-FE  
Chibu Ofoegbu- PJM  
Dave Egan- PJM

File

***Generation Interconnection  
System Impact Study Report***

***For***

***PJM Generation Interconnection Request  
Queue Position AD1-113***

***“Raritan River 230 kV”***

***35 MWE, 75 MWC Uprate***

***September 2018***

## Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

## Changes from Feasibility Study:

Per request of the Interconnection Customer, the increase in Maximum Facility Output (MFO)/Energy was reduced from 75 MW to 35 MW for the AD1-113 project. PJM determined this change was not a material modification. The Capacity value request did not change. See the updated values in the table below in the “General” section.

## General

**CPV Shore, LLC**, the Interconnection Customer (IC), has proposed an increase in output of their existing Woodbridge Energy Center natural gas generating facility located at 1070 Riverside Drive, Keasbey, New Jersey (Attachment 1). The uprate will increase the maximum facility output (MFO) of the facility by **35 MW**, for a total plant MFO of 820 MW. The uprate will also increase the Capacity value by **75 MW**, for a total of 800 MW being recognized by PJM as Capacity. (See the summary table below.) The proposed in-service date for this project is **December 1, 2018**. **This study does not imply a Jersey Central Power and Light (JCP&L) commitment to this in-service date.**

| Queue Number | MFO (MW) and Winter Energy | Capacity (MW) and Summer Energy |
|--------------|----------------------------|---------------------------------|
| W4-009       | 725                        | 725                             |
| X4-005       | 785 (60 MW increase)       | 725 (0 MW increase)             |

|              |                      |                      |
|--------------|----------------------|----------------------|
| AD1-113      | 820 (35 MW increase) | 800 (75 MW increase) |
| <b>Total</b> | <b>820</b>           | <b>800</b>           |

PJM studied the proposed request and evaluated it for compliance with reliability criteria for summer peak conditions in 2021.

## Point of Interconnection

**AD1-113 “Raritan River 230 kV”** uprate project will be an increase to the existing Woodbridge Energy Center natural gas generation plant which interconnects to the JCP&L transmission system at the Raritan River 230 kV bus. As defined by CPV Shore, LLC and shown on Attachment 2, the connection point for this project will be accomplished by utilizing the existing 230 kV point of interconnection for the Woodbridge generation plant at JCP&L’s Raritan River substation.

Attachment 2 shows a one-line diagram of the connection facilities that exist for the (AD1-113) generation project. As indicated, it was studied as a 820 MW injection to the Jersey Central 230 kV system at the existing 230 kV connection for the Woodbridge generation plant at JCP&L’s Raritan River substation. The JCP&L facilities required to be upgraded for the primary direct connection, non-direct connection, and network upgrades of the generation project and the associated cost estimate are outlined below.

## Cost Summary

The AD1-113 “Raritan River 230 kV” project will be responsible for the following costs:

| <b>Description</b>                             | <b>Total Cost</b> |
|------------------------------------------------|-------------------|
| Attachment Facilities                          | \$ 0              |
| Direct Connection Network Upgrades             | \$ 0              |
| Non Direct Connection Network Upgrades         | \$ 168,800        |
| New System Upgrades                            | \$ 0              |
| Contribution to Previously Identified Upgrades | \$ 0              |
| <b>Total Costs</b>                             | <b>\$ 168,800</b> |

The transmission and substation costs given above exclude any applicable state or federal taxes. If at a future date Federal CIAC (contribution in aid of construction) taxes are deemed necessary by the IRS for this project, JCP&L shall be reimbursed by the Interconnection Customer for such taxes. JPCL estimates the tax, if applicable, would be approximately **\$26,000**.

The required Attachment Facilities, Direct Connection, and Non-Direct Connection work for the interconnection of the AD1-113 generation uprate project to the JCP&L Transmission System is detailed in the following sections. The associated one-line with the generation project is shown in Attachment 2.

Note that the FE findings were made from a conceptual review of this project. A more detailed review of the connection facilities and their cost will be identified in the System Impact and Facilities Studies. Further note that the cost estimate data contained in this document should be considered high level estimates since it was produced without a detailed engineering review. The applicant will be responsible for the actual cost of construction. FE herein reserves the right to return to any issues in this document and, upon appropriate justification, request additional monies to complete any reinforcements to the transmission systems.

## Attachment Facilities

There is no Attachment Facilities work required for this project.

## Direct Connection Cost Estimate

There is no Direct Connection scope of work required for this project.

## Non-Direct Connection Cost Estimate

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

| Description                                                                                                                        | Total Cost        |
|------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Make relay setting changes on the 230kV line terminals. @ Raritan River SS<br><i>(PJM Network Upgrade Number n5752)</i>            | \$ 42,200         |
| Make relay setting changes on the 230kV line terminals to Raritan River. @ Parlin SS<br><i>(PJM Network Upgrade Number n5753)</i>  | \$ 42,200         |
| Make relay setting changes on the 230kV line terminals to Raritan River. @ Red Oak SS<br><i>(PJM Network Upgrade Number n5754)</i> | \$ 42,200         |
| Make relay setting changes on the 230kV line terminals to Raritan River. @ Werner SS<br><i>(PJM Network Upgrade Number n5755)</i>  | \$ 42,200         |
| <b>Total Non-Direct Connection Facility Costs</b>                                                                                  | <b>\$ 168,800</b> |

The total Non-Direct Connection cost estimate for the AD1-113 project is approximately **\$168,800**. This work is to perform required relay setting changes at JCP&L's Raritan River, Parlin, Red Oak, and Werner Substations.

## Schedule

Based on the extent of the JCP&L primary Non-Direct Connection work required to support the AD1-113 generation project, it is expected to take a **minimum of 7 months** from the date of a fully executed Interconnection Construction Service Agreement to complete the installation subject to market conditions and vendor lead times. This includes the requirement for the Interconnection Customer to make a preliminary payment to JCP&L which funds the construction of the Non-Direct Connection facilities work. It assumes that there will be no environmental or permitting issues to implement the upgrades required for this uprate project and that all system outages will be allowed when requested.

## JCP&L Analysis

### Power Flow Analysis

PJM performed a power flow analysis of the transmission system using a 2021 summer peak load flow model and a 2021 winter peak load flow model and the results were verified by FE. Additionally, FE performed an analysis of its underlying transmission <100 kV system. At the Primary POI, the AD1-113 project does not contribute to overloads on the FE transmission system. However, it contributes to an overload on the CPV generator lead line as shown in Attachment 2a. The estimated costs of system reinforcements necessary to mitigate this overload is not provided as the affected facility is owned by CPV Shore, LLC.

### Power Factor Requirements

The existing 785 MW portion of the Customer Facility shall retain its existing ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.90 lagging (supplying VARs) measured at the generator's terminals. The increase of 35 MW to the Customer Facilities associated with the AD1-113 project shall be designed with the ability to maintain a Power Factor of at least 1.0 (unity) to 0.90 lagging (supplying VARs) measured at the generator's terminals.

FE performed a reactive capability analysis of the proposed incremental increase. The following table(s) show the reactive power requirements and capabilities for both the existing and additional output.

| Queue                                                                | Size (MW)  | Leading PF  | Lagging PF  | Leading MVARs       | Lagging MVARs         |
|----------------------------------------------------------------------|------------|-------------|-------------|---------------------|-----------------------|
| W4-009/X4-005                                                        | 785        | 0.95        | 0.90        | -258                | 380                   |
| AD1-113                                                              | 35         | 1.0         | 0.90        | 0                   | 17                    |
| <b>Total</b>                                                         | <b>820</b> | <b>0.95</b> | <b>0.90</b> | <b>-258</b>         | <b>397</b>            |
| <b>Estimated Reactive Capability at 820 MW (IC provided D-Curve)</b> |            |             |             | <b>-280</b>         | <b>392</b>            |
| <b>MVAR Surplus/Deficiency</b>                                       |            |             |             | <b>22 (Surplus)</b> | <b>-5 (Deficient)</b> |

Note that a further conclusion of this study is that it will be mandatory for the Woodbridge generation plant to retain the power factor requirements stated in the Interconnection Service Agreements for the existing generation. Should AD1-113 fail to provide dynamic reactive capability from the generation project for any reason once interconnected, the FE and/or PJM Dispatchers may need to take action to curtail its output to prevent non-compliance with the power factor criteria.

### **Short Circuit and Dynamics Analysis**

In accordance with the RTEP process, short circuit and dynamics analysis will not be conducted by PJM since the (AD1-113) generation project capacity increase request has no change to the electrical characteristics of the CPV Shore, LLC generating facility as indicated by the developer.

### **Stability Analysis**

A dynamic stability analysis was completed by PJM and the results were reviewed by FE. The results of the stability will be provided to the IC in PJM's System Impact Study report. There were no stability concerns identified for the faults studied.

### **System Protection Analysis**

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

### **Compliance Issues**

The proposed Customer Facilities must be designed in accordance with FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated 230 kV circuit breaker to protect the AD1-113 generator lead line. A single circuit breaker must be used to protect this line; if the project has several GSU transformers, the individual GSU transformer breakers cannot be used to protect this line.
2. The purchase and installation of the minimum required FE generation interconnection relaying and control facilities. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition ("SCADA") equipment to provide information in a compatible format to the FE Transmission System Control Center.

4. Compliance with the FE and PJM generator power factor and voltage control requirements.
5. The execution of a back-up service agreement to serve the customer load supplied from the AD1-113 generation project metering point when the units are out-of-service. This assumes the intent of the IC is to net the generation with the load.

The IC will also be required to meet all PJM, ReliabilityFirst, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and ReliabilityFirst audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the FE system.

## **CPV Shore, LLC Requirements**

CPV Shore, LLC will be responsible for meeting all criteria as specified in the applicable sections of the FE “Requirements for Transmission Connected Facilities” document including compliance with the FE and PJM generator power factor and voltage control requirements.

<http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>

The above requirements are in addition to any metering or other requirements imposed by PJM.

## **Revenue Metering and SCADA Requirements**

### **PJM Requirements**

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC’s generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

### **JCP&L Requirements**

The IC will be required to comply with all FE revenue metering requirements for generation interconnection customers which can be found in FE’s “Requirements for Transmission Connected Facilities” document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>.

## Network Impacts

The Queue Project AD1-113 was evaluated as a 35 MW MFO (Capacity 75 MW) uprate injection into the Raritan River 230 kV bus. Project AD1-113 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AD1-113 was studied with a commercial probability of 53%. Potential network impacts were as follows:

### Summer Peak Analysis - 2021

#### Generator Deliverability

*(Single or N-1 contingencies for the Capacity portion only of the interconnection)*

None

#### Multiple Facility Contingency

*(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)*

None

#### Contribution to Previously Identified Overloads

*(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)*

None

#### Steady-State Voltage Requirements

*(Results of the steady-state voltage studies should be inserted here)*

None

#### Stability and Reactive Power Requirement for Low Voltage Ride Through

*(Summary of the VAR requirements based upon the results of the dynamic studies)*

No mitigations were found to be required. (See Attachment 3 for Stability Study Report.)

#### Short Circuit

*(Summary of impacted circuit breakers)*

None

### Affected System Analysis & Mitigation

#### NYISO Impacts:

NYISO Impacts to be determined during later study phases (as applicable).

**Delivery of Energy Portion of Interconnection Request**

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

None

**Light Load Analysis - 2021**

No light load thermal impacts.

**Winter Analysis – 2021**

No winter thermal impacts.

**System Reinforcements**

**Short Circuit**

*(Summary form of Cost allocation for breakers will be inserted here if any)*

None

**Stability and Reactive Power Requirement**

*(Results of the dynamic studies should be inserted here)*

None

**Summer Peak Load Flow Analysis Reinforcements**

**New System Reinforcements**

*(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)*

None

**Contribution to Previously Identified System Reinforcements**

*(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)*

*(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)*

None

**Light Load Load Flow Analysis Reinforcements**

**New System Reinforcements**

*(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)*

None

**Contribution to Previously Identified System Reinforcements**

*(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)*  
*(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)*

None

**Winter Load Flow Analysis Reinforcements**

**New System Reinforcements**

*(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)*

None

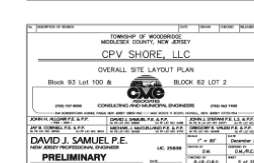
**Contribution to Previously Identified System Reinforcements**

*(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)*  
*(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)*

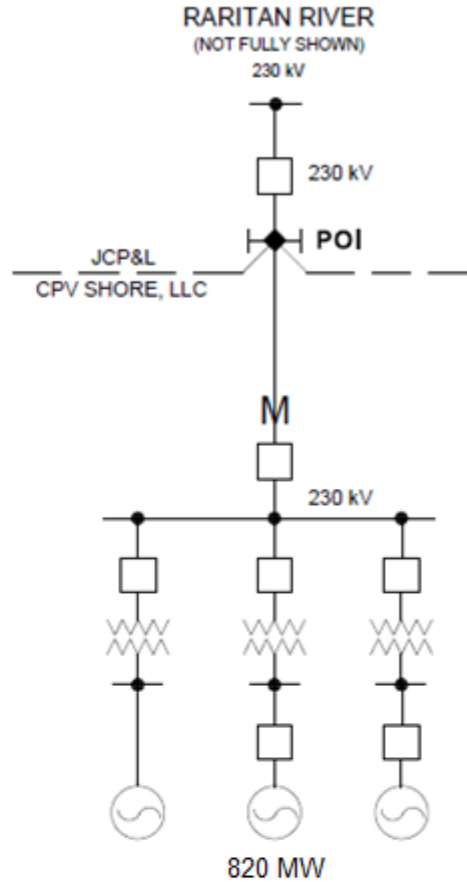
None

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## Project Location



## Attachment 2: (AD1-113) Generation Project Planning Interconnection Single Line Diagram



◆ = POI (POINT OF INTERCONNECTION) LOCATED AT JERSEY CENTRAL POWER & LIGHT (JCP&L) SUBSTATION DEAD-END STRUCTURE, WHERE CPV SHORE, LLC TRANSMISSION LINE TERMINATES.

M = REVENUE METERING FOR CPV SHORE, LLC IS OWNED, OPERATED, AND MAINTAINED BY CPV SHORE, LLC.

|                                                          |                    |                                                                                                                                          |                                 |
|----------------------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>FirstEnergy</b><br>Energy Delivery Technical Services |                    | TITLE<br>JERSEY CENTRAL POWER & LIGHT (JCP&L) RARITAN SUBSTATION 230 kV<br>INTERCONNECTION TO CPV SHORE, LLC OWNED GENERATION FACILITIES |                                 |
| BY<br>J.L.R.                                             | DATE<br>07/26/2018 | APPROVED<br>-                                                                                                                            | POI-JC-AD1-113.dwg<br>REV.<br>- |

**Attachment 3: (AD1-113) Generation Project  
Dynamic Stability Analysis Results**

**AD1-113  
System Impact Study**

**Dynamic Simulation Analysis**

Version 0

April 6, 2018

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## Executive Summary

Generator Interconnection Request AD1-113 project is for a 35 MW Maximum Facility Output (MFO) uprate to the W4-009 interconnection request. After the uprate, AD1-113 is for a 820MW MFO 2 Combustine Turbines and 1 Steam Turbines combine cycle generating facility with the Point of Interconnection (POI) on Raritan River 230kV station in JCPL system located in Middlesex County, New Jersey.

This report describes a dynamic simulation analysis of AD1-113 as part of the overall system impact study.

The load flow scenario for the analysis was based on the RTEP 2021 Light Load case, modified to include applicable queue projects. AD1-113 has been dispatched online at maximum power output, with regulating Raritan River 230kV bus voltage at 1.008 p.u., which is consistent with the default generator reference voltage specified in PJM Manual 03 Transmission Operations Section 3.3.3 for generator connections to the PJM 230 kV system.

The AD1-113 queue project was tested for compliance with NERC, PJM and other applicable criteria. The range of contingencies evaluated was limited to that necessary to assess compliance and each was limited to a 20-second simulation time period.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are not applicable.

Other applicable criteria tested include:

1. Transmission Owner (TO) specific criteria
2. Other criteria

---

The system was tested for a system intact condition and the fault types listed above. Specific fault descriptions and breaker clearing times used for this study are provided in the result table.

For all simulations, the queue project under study along with the rest of the PJM system were required to maintain synchronism and with all states returning to an acceptable new condition following the disturbance.

For the fault contingencies tested on the 2021 light load case:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 3%.
- b) The AD1-113 generator was able to ride through all faults (except for faults where protective action trips a generator(s)).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

No mitigations were found to be required.

---

## **1. Introduction**

Generator Interconnection Request AD1-113 project is for a 35 MW Maximum Facility Output (MFO) uprate to the W4-009 interconnection request. After the uprate, AD1-113 is for a 820MW MFO 2 Combustion Turbines and 1 Steam Turbines combine cycle generating facility with the Point of Interconnection (POI) on Raritan River 230kV station in JCPL system located in Middlesex County, New Jersey.

This analysis is effectively a screening study to determine whether the addition of AD1-113 will meet the dynamic requirements of the NERC, PJM and Transmission Owner reliability standards.

In this report the AD1-113 project and how it is proposed to be connected to the grid are first described, followed by a description of how the project is modeled in this study. The fault cases are then described and analyzed, and lastly a discussion of the results is provided.

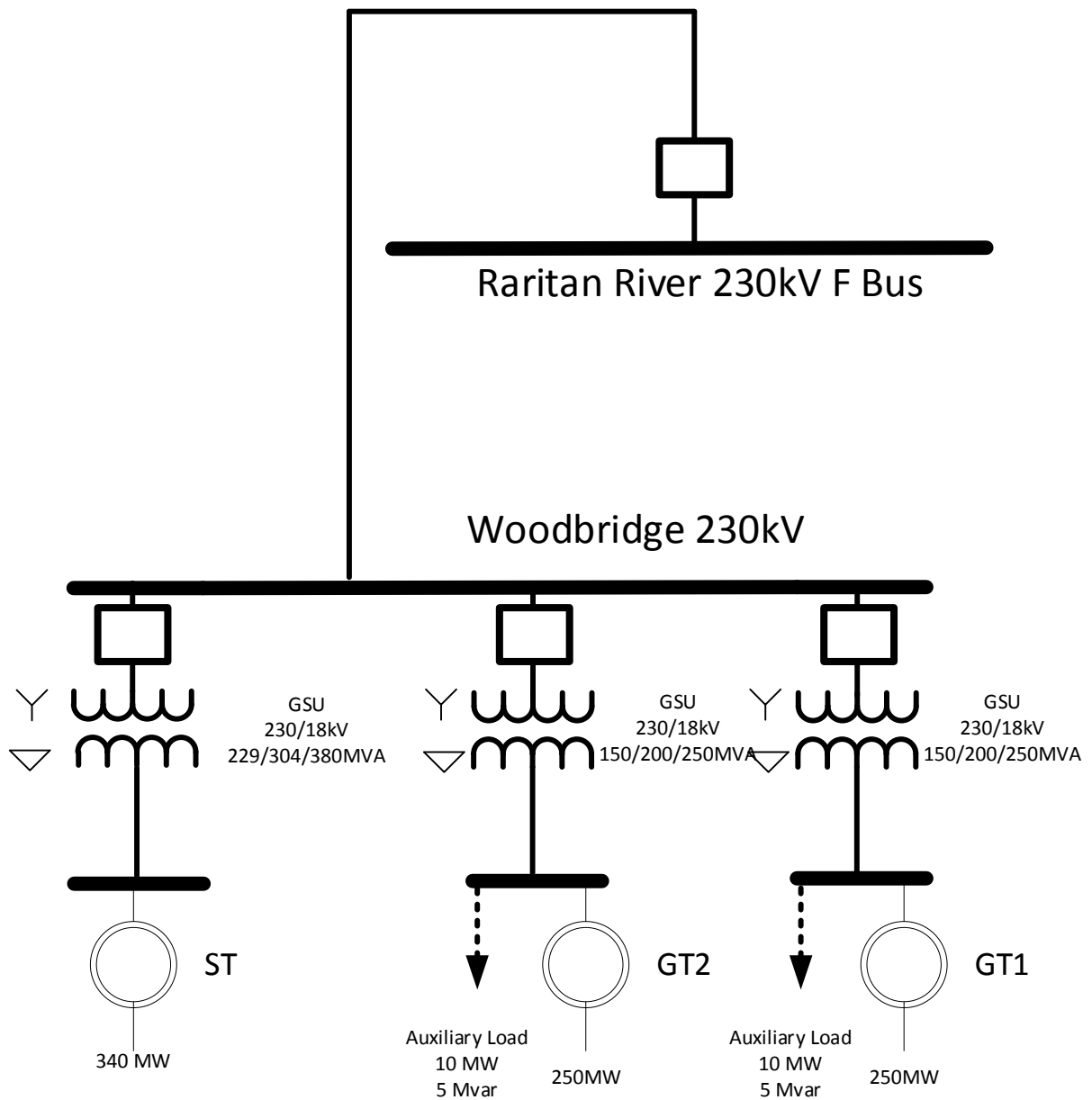
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## **2. Description of Project**

Generator Interconnection Request AD1-113 project is for a 35 MW Maximum Facility Output (MFO) uprate to the W4-009 interconnection request. After the uprate, AD1-113 is for a 820MW MFO 2 Combustine Turbines and 1 Steam Turbines combine cycle generating facility with the Point of Interconnection (POI) on Raritan River 230kV station in JCPL system located in Middlesex County, New Jersey.

Figure 1 shows the simplified one-line diagram of the AD1-113 loadflow model. Table 1 lists the parameters given in the impact study data and the corresponding parameters of the AD1-113 loadflow model.

The dynamic models for AD1-113 plant are based on the PSS\E standard model with parameters supplied by the developer.



**Figure 1: AD1-113 Plant Model**

**Table 1: AD1-113 Plant Model**

|            | Impact Study Data                                         | Model                                 |
|------------|-----------------------------------------------------------|---------------------------------------|
| Generators | 2 X 250MW GT                                              | 2 × 250 MW GE GT                      |
|            | MVA base = 262 MVA                                        | Pgen            250 MW                |
|            | Vt = 18 kV                                                | Pmax            250 MW                |
|            |                                                           | Pmin            0 MW                  |
|            | Unsaturated sub-transient reactance = 0.205 pu @ MVA base | Qmax            135 MVA <sub>r</sub>  |
|            |                                                           | Qmin            -90 MVA <sub>r</sub>  |
|            |                                                           | Mbase           262 MVA               |
|            |                                                           | Zsorce           j0.205 pu @ Mbase    |
|            | 1 × 340 MW ST                                             | 1 × 340 MW ST                         |
|            | MVA base = 385 MVA                                        | Pgen            340 MW                |
|            | Vt = 18 kV                                                | Pmax            340 MW                |
|            |                                                           | Pmin            0 MW                  |
|            | Unsaturated sub-transient reactance = j0.25 pu @ MVA base | Qmax            165 MVA <sub>r</sub>  |
|            |                                                           | Qmin            -135 MVA <sub>r</sub> |
|            |                                                           | Mbase           385 MVA               |
|            |                                                           | Zsorce           j0.25 pu @ Mbase     |

|                    | Impact Study Data                          | Model                                      |
|--------------------|--------------------------------------------|--------------------------------------------|
| GSU transformer(s) | GT1 GSU 230/18 kV two winding transformer  | GT1 GSU 230/18 kV two winding transformer  |
|                    | Rating = 150/200/250 MVA                   | Rating = 150/200/250 MVA                   |
|                    | Transformer base = 150 MVA                 | Transformer base = 150 MVA                 |
|                    | Impedance = 0.00142 + j0.0934pu @ MVA base | Impedance = 0.00142 + j0.0934pu @ MVA base |
|                    | Number of taps = N/A                       | Number of taps = 5                         |
|                    | Tap step size = N/A                        | Tap step size = 2.5%                       |
|                    | GT2 GSU 230/18 kV two winding transformer  | GT2 GSU 230/18 kV two winding transformer  |
|                    | Rating = 150/200/250 MVA                   | Rating = 150/200/250 MVA                   |
|                    | Transformer base = 150 MVA                 | Transformer base = 150 MVA                 |
|                    | Impedance = 0.00142 + j0.0934pu @ MVA base | Impedance = 0.00142 + j0.0934pu @ MVA base |
|                    | Number of taps = N/A                       | Number of taps = 5                         |
|                    | Tap step size = N/A                        | Tap step size = 2.5%                       |
|                    | Page 9                                     |                                            |
|                    | ST GSU 230/18kV tho winding transformer    | ST GSU 230/18kV tho winding transformer    |

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|                 | <b>Impact Study Data</b>               | <b>Model</b>                                            |
|-----------------|----------------------------------------|---------------------------------------------------------|
| Auxiliary load  | 2 x 10 MW, 5 MVA <sub>r</sub>          | 2 x 10 MW, 5 MVA <sub>r</sub>                           |
| Station load    | N/A                                    | N/A                                                     |
| Attachment line | Not Provided, use as built in the case | R = 0.00208 p.u.<br>X = 0.02288 p.u.<br>B = 0.0074 p.u. |

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### 3. Loadflow and Dynamics Case Setup

The dynamics simulation analysis was carried out using PSS/E Version 33.7.

The load flow scenario and fault cases for this study are based on PJM's Regional Transmission Planning Process<sup>1</sup>.

The selected load flow scenario is the RTEP 2021 light load case with the following modifications:

- a) Addition of all applicable queue projects prior to AD1-113.
- b) Addition of AD1-113 queue project.
- c) Removal of withdrawn and subsequent queue projects in the vicinity of AD1-113.
- d) Dispatch of units in the PJM system to maintain slack generators within limits.

The AD1-113 initial conditions are listed in Table 2, indicating maximum power output with regulating Raritan River 230kV bus voltage at 1.008 p.u., which is consistent with the default generator reference voltage specified in PJM Manual 03 Transmission Operations Section 3.3.3 for generator connections to the PJM 230 kV system.

**Table 2: AD1-113 machine initial conditions**

| <b>Bus</b> | <b>Name</b> | <b>Unit</b> | <b>PGEN</b> | <b>QGEN</b> | <b>ETERM</b> | <b>POI Voltage</b> |
|------------|-------------|-------------|-------------|-------------|--------------|--------------------|
| 934841     | AD1-113 CT1 | 1           | 250         | 17.4        | 0.99         | 1.008              |
| 934842     | AD1-113 CT2 | 1           | 250         | 17.4        | 0.99         | 1.008              |
| 934843     | AD1-113 ST  | 1           | 250         | 17.4        | 0.99         | 1.008              |

Generation within the PJM500 system (area 225 in the PSS/E case) and within the vicinity of AD1-113 has been dispatched online at maximum output (P<sub>MAX</sub>). The dispatch of generation in the vicinity of AD1-113 is given in Attachment 3.

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<sup>1</sup> Manual 14B: PJM Region Transmission Planning Process, Rev 33, May 5 2016, Attachment G : PJM Stability, Short Circuit, and Special RTEP Practices and Procedures.

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## 4. Fault Cases

Tables 3 listed the contingencies and results that were studied, with representative worst case total clearing times provided by PJM. Each contingency was studied over a 20 second simulation time interval.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are not applicable.

Other applicable criteria tested include:

1. Transmission Owner (TO) specific criteria
2. Other criteria

The system was tested for a system intact condition and the fault types listed above.

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## 5. Evaluation Criteria

This study is focused on AD1-113, along with the rest of the PJM system, maintaining synchronism and having all states return to an acceptable new condition following the disturbance. The recovery criteria applicable to this study are as per PJM's Regional Transmission Planning Process and Transmission Owner criteria:

- a) The system with AD1-113 included is transiently stable and post-contingency oscillations should be positively damped with a damping margin of at least 3%
- b) The AD1-113 is able to ride through faults (except for faults where protective action trips AD1-113).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

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## 6. Summary of Results

Plots from the dynamic simulations are provided in Attachment 4, with results summarized in Table 3.

For the studied fault contingencies tested on the 2021 light load case:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 3%.
- b) The AD1-113 generator was able to ride through all faults (except for faults where protective action trips a generator(s)).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

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## **7. System Reinforcement Requirements**

None.

**Table 3: Contingency List and Stability Study Results****P0: Stead State**

| <b>Fault ID</b> | <b>Duration</b>         |
|-----------------|-------------------------|
| P0.00           | Steady State 20 sec run |

**P1: Three Phase Faults with normal clearing**

| <b>Fault ID</b> | <b>Fault description</b>                                                                        | <b>Clearing Time Normal &amp; HSR (Cycles)</b> | <b>Results</b> |
|-----------------|-------------------------------------------------------------------------------------------------|------------------------------------------------|----------------|
| P1.00           | 3ph @ Raritan River – Woodbridge 230kV, normal clear loss of AD1-113                            | 5                                              | Stable         |
| P1.01           | 3ph @ Raritan River Sayerville GSU, normal clear loss of Raritan CT 1-4                         | 5                                              | Stable         |
| P1.02           | 3ph @ Raritan River – Red Oak – South River Junction, normal clear loss of Red Oak unit 2 and 3 | 5                                              | Stable         |
| P1.03           | 3ph @ Raritan River – Red Oak – South River Junction, normal clear loss of Red Oak unit 1 and 4 | 5                                              | Stable         |
| P1.04           | 3ph @ Raritan River – NJ Steel – Werner 230kV, normal clear                                     | 5                                              | Stable         |
| P1.05           | 3ph @ Raritan River – Kilmer – Lake Nelson 230kV ckt 1, normal clear                            | 5                                              | Stable         |
| P1.06           | 3ph @ Raritan River – Kilmer – Lake Nelson 230kV ckt 2, normal clear                            | 5                                              | Stable         |
| P1.07           | 3ph @ Raritan River 230kV, normal clear loss of Raritan River 230/115kV TF 17                   | 5                                              | Stable         |
| P1.08           | 3ph @ Raritan River 230kV, normal clear loss of Raritan River 230/115kV TF 13                   | 5                                              | Stable         |
| P1.09           | 3ph @ Raritan River – Neptune Converter 230kV, normal clear                                     | 5                                              | Stable         |
| P1.10           | 3ph @ Raritan River 115kV, normal clear loss of Raritan River 230/115kV TF 17                   | 7                                              | Stable         |
| P1.11           | 3ph @ Raritan River 115kV, normal clear loss of Raritan River 230/115kV TF 13                   | 7                                              | Stable         |
| P1.12           | 3ph @ Raritan River 115kV, normal clear loss of Raritan River 115/34.5kV TF 4                   | 7                                              | Stable         |
| P1.13           | 3ph @ Raritan River 115kV, normal clear loss of Raritan River 115/34.5kV TF 14                  | 7                                              | Stable         |
| P1.14           | 3ph @ Raritan River – Werner 115kV, normal clear                                                | 7                                              | Stable         |

| <b>Fault ID</b> | <b>Fault description</b>                                                                              | <b>Clearing Time Normal &amp; HSR (Cycles)</b> | <b>Results</b> |
|-----------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------|----------------|
| P1.15           | 3ph @ Raritan River – Deep Run 115kV ckt 1, normal clear                                              | 7                                              | Stable         |
| P1.16           | 3ph @ Raritan River – Deep Run 115kV ckt 2, normal clear                                              | 7                                              | Stable         |
| P1.17           | 3ph @ Lake Nelson – Kilmer – Raritan River 230kV ckt 1, normal clear                                  | 5                                              | Stable         |
| P1.18           | 3ph @ Lake Nelson – Kilmer – Raritan River 230kV ckt 2, normal clear                                  | 5                                              | Stable         |
| P1.19           | 3ph @ Lake Nelson – Green Brook 230kV, normal clear                                                   | 5                                              | Stable         |
| P1.20           | 3ph @ Lake Nelson – Middlesex 230kV, normal clear                                                     | 5                                              | Stable         |
| P1.21           | 3ph @ Green Brook – Lake Nelson 230kV, normal clear                                                   | 5                                              | Stable         |
| P1.22           | 3ph @ Green Brook – Gillette 230kV, normal clear                                                      | 5                                              | Stable         |
| P1.23           | 3ph @ Middlesex – Lake Nelson 230kV, normal clear                                                     | 5                                              | Stable         |
| P1.24           | 3ph @ Middlesex – Bridgewater 230kV, normal clear                                                     | 5                                              | Stable         |
| P1.25           | 3ph @ Middlesex – Green Brook – Gillette 230kV, normal clear                                          | 5                                              | Stable         |
| P1.26           | 3ph @ South River Junction – Red Oak – Raritan River 230kV, normal clear loss of Red Oak unit 2 and 3 | 5                                              | Stable         |
| P1.27           | 3ph @ South River Junction – Red Oak – Raritan River 230kV, normal clear loss of Red Oak unit 1 and 4 | 5                                              | Stable         |
| P1.28           | 3ph @ South River Junction – AC1-029 230kV, normal clear                                              | 5                                              | Stable         |
| P1.29           | 3ph @ South River Junction – South River 230kV, normal clear                                          | 5                                              | Stable         |
| P1.30           | 3ph @ South River Junction – Parlin 230kV, normal clear                                               | 5                                              | Stable         |
| P1.31           | 3ph @ South River Junction – Atlantic 230kV, normal clear                                             | 5                                              | Stable         |
| P1.32           | 3ph @ AC1-029 POI – South River 230kV, normal clear                                                   | 5                                              | Stable         |
| P1.33           | 3ph @ AC1-029 POI – AC1-029 Main 230kV, normal clear loss of AC1-029                                  | 5                                              | Stable         |
| P1.34           | 3ph @ AC1-029 POI – South River Junction 230kV, normal clear                                          | 5                                              | Stable         |
| P1.35           | 3ph @ South River – South River Junction 230kV, normal clear                                          | 5                                              | Stable         |
| P1.36           | 3ph @ South River – AC1-029 POI 230kV, normal clear                                                   | 5                                              | Stable         |
| P1.37           | 3ph @ South River 230kV, normal clear loss of South River Unit 1                                      | 5                                              | Stable         |
| P1.38           | 3ph @ South River 230kV, normal clear loss of South River Unit 2 and 3                                | 5                                              | Stable         |
| P1.39           | 3ph @ Parlin – South River Junction 230kV, normal clear                                               | 5                                              | Stable         |

| <b>Fault ID</b> | <b>Fault description</b>                                                                 | <b>Clearing Time Normal &amp; HSR (Cycles)</b> | <b>Results</b> |
|-----------------|------------------------------------------------------------------------------------------|------------------------------------------------|----------------|
| P1.40           | 3ph @ Parlin – Williams Gas – Freneau 230kV, normal clear                                | 5                                              | Stable         |
| P1.41           | 3ph @ Parlin 230kV, normal clear loss of Parlin unit 1 and 2, Parlin 230kV load (Dupont) | 5                                              | Stable         |
| P1.42           | 3ph @ Parlin 230kV, normal clear loss of Parlin unit 3 and 4                             | 5                                              | Stable         |

#### **P4: SLG Stuck Breaker (SB) Faults at Backup Clearing**

| <b>Fault ID</b> | <b>Fault description</b>                                                                                                                                                                                              | <b>Clearing Time Near &amp; Remote (Cycles)</b> | <b>Results</b> |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------|
| P4.01           | SLG @ Raritan River 230kV, normal clear loss of Sayerville units, Raritan River 230/115kV TF 17, SB @ Raritan River, delayed clear loss of Raritan River – Neptune Converter 230kV                                    | 5 / 16                                          | Stable         |
| P4.02           | SLG @ Raritan River 230kV, normal clear loss of Sayerville units, Raritan River 230/115kV TF 17, SB @ Raritan River, delayed clear loss of Raritan River – Kilmer – Lake Nelson 230kV ckt 1                           | 5 / 16                                          | Stable         |
| P4.03           | SLG @ Raritan River 230kV, normal clear loss of Sayerville units, Raritan River 230/115kV TF 17, SB @ Raritan River, delayed clear loss of Raritan River – Kilmer – Lake Nelson 230kV ckt 2                           | 5 / 16                                          | Stable         |
| P4.04           | SLG @ Raritan River 230kV, normal clear loss of Sayerville units, Raritan River 230/115kV TF 17, SB @ Raritan River, delayed clear loss of Raritan River – NJ Steel – Werner 230kV                                    | 5 / 16                                          | Stable         |
| P4.05           | SLG @ Raritan River 230kV, normal clear loss of Sayerville units, Raritan River 230/115kV TF 17, SB @ Raritan River, delayed clear loss of Raritan River – Red Oak – South River Junction 230kV, Red Oak unit 1 and 4 | 5 / 16                                          | Stable         |
| P4.06           | SLG @ Raritan River 230kV, normal clear loss of Sayerville units, Raritan River 230/115kV TF 17, SB @ Raritan River, delayed clear loss of Raritan River – Red Oak – South River Junction 230kV, Red Oak unit 2 and 3 | 5 / 16                                          | Stable         |
| P4.07           | SLG @ Raritan River 230kV, normal clear loss of Raritan River – Woodbridge 230kV, AD1-113, Raritan River 230/115kV TF 13, SB @ Raritan River, delayed clear loss of Raritan River – Neptune Converter 230kV           | 5 / 16                                          | Stable         |
| P4.08           | SLG @ Raritan River 230kV, normal clear loss of Raritan River – Woodbridge 230kV, AD1-113, Raritan River 230/115kV TF 13, SB @ Raritan River, delayed clear loss of Raritan River – Kilmer – Lake Nelson 230kV ckt 1  | 5 / 16                                          | Stable         |

| <b>Fault ID</b> | <b>Fault description</b>                                                                                                                                                                                                                       | <b>Clearing Time Near &amp; Remote (Cycles)</b> | <b>Results</b> |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------|
| P4.09           | SLG @ Raritan River 230kV, normal clear loss of Raritan River – Woodbridge 230kV, AD1-113, Raritan River 230/115kV TF 13, SB @ Raritan River, delayed clear loss of Raritan River – Kilmer – Lake Nelson 230kV ckt 2                           | 5 / 16                                          | Stable         |
| P4.10           | SLG @ Raritan River 230kV, normal clear loss of Raritan River – Woodbridge 230kV, AD1-113, Raritan River 230/115kV TF 13, SB @ Raritan River, delayed clear loss of Raritan River – NJ Steel – Werner 230kV                                    | 5 / 16                                          | Stable         |
| P4.11           | SLG @ Raritan River 230kV, normal clear loss of Raritan River – Woodbridge 230kV, AD1-113, Raritan River 230/115kV TF 13, SB @ Raritan River, delayed clear loss of Raritan River – Red Oak – South River Junction 230kV, Red Oak unit 1 and 4 | 5 / 16                                          | Stable         |
| P4.12           | SLG @ Raritan River 230kV, normal clear loss of Raritan River – Woodbridge 230kV, AD1-113, Raritan River 230/115kV TF 13, SB @ Raritan River, delayed clear loss of Raritan River – Red Oak – South River Junction 230kV, Red Oak unit 2 and 3 | 5 / 16                                          | Stable         |
| P4.13           | SLG @ Raritan River 230kV, normal clear loss of Raritan River 230/115kV TF 17, SB @ Raritan River, delayed clear loss of Sayerville units                                                                                                      | 5 / 16                                          | Stable         |
| P4.14           | SLG @ Raritan River 230kV, normal clear loss of Sayerville units, SB @ Raritan River, delayed clear loss of Raritan River 230/115kV TF 17                                                                                                      | 5 / 16                                          | Stable         |
| P4.15           | SLG @ Raritan River 230kV, normal clear loss of Raritan 230/115kV TF 13, SB @ Raritan River, delayed clear loss of Raritan River – Woodbridge 230kV, AD1-113                                                                                   | 5 / 16                                          | Stable         |
| P4.16           | SLG @ Raritan River 230kV, normal clear loss of Raritan River – Woodbridge 230kV, AD1-113, SB @ Raritan River, delayed clear loss of Raritan 230/115kV TF 13                                                                                   | 5 / 16                                          | Stable         |
| P4.17           | SLG @ Raritan River 115kV, normal clear loss of Raritan River 115/34.5kV TF14, SB @ Raritan River 115kV, delayed clear loss of Raritan River 230/115kV TF 17                                                                                   | 7 / 20                                          | Stable         |
| P4.18           | SLG @ Raritan River 115kV, normal clear loss of Raritan River 115/34.5kV TF14, SB @ Raritan River 115kV, delayed clear loss of Raritan River – Deep Run 115kV ckt 1                                                                            | 7 / 20                                          | Stable         |
| P4.19           | SLG @ Raritan River 115kV, normal clear loss of Raritan River 115/34.5kV TF4, SB @ Raritan River 115kV, delayed clear loss of Raritan River 230/115kV TF 17                                                                                    | 7 / 20                                          | Stable         |
| P4.20           | SLG @ Raritan River 115kV, normal clear loss of Raritan River 115/34.5kV TF4, SB @ Raritan River 115kV, delayed clear loss of Raritan River 230/115kV TF 13                                                                                    | 7 / 20                                          | Stable         |
| P4.21           | SLG @ Raritan River 115kV, normal clear loss of Raritan River – Deep Run ckt 2, SB @ Raritan River 115kV, delayed clear loss of Raritan River – Deep Run 115kV ckt 1                                                                           | 7 / 20                                          | Stable         |

| <b>Fault ID</b> | <b>Fault description</b>                                                                                                                                                                                                                                                 | <b>Clearing Time Near &amp; Remote (Cycles)</b> | <b>Results</b> |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------|
| P4.22           | SLG @ Raritan River 115kV, normal clear loss of Raritan River – Deep Run ckt 2, SB @ Raritan River 115kV, delayed clear loss of Raritan River – Werner 115kV                                                                                                             | 7 / 20                                          | Stable         |
| P4.23           | SLG @ Lake Nelson 230kV, normal clear loss of Lake Nelson – Green Brook 230kV, SB @ Lake Nelson, delayed clear loss of Lake Nelson – Kilmer – Raritan River 230kV ckt 2                                                                                                  | 5 / 17                                          | Stable         |
| P4.24           | SLG @ Lake Nelson 230kV, normal clear loss of Lake Nelson – Middlesex 230kV, SB @ Lake Nelson, delayed clear loss of Lake Nelson – Kilmer – Raritan River 230kV ckt 1                                                                                                    | 5 / 17                                          | Stable         |
| P4.25           | SLG @ South River Junction 230kV, normal clear loss of South River Junction – Red Oak – Raritan River 230kV line, Red Oak unit 2 and 3, SB @ South River Junction, delayed clear loss of South River Junction – Red Oak – Raritan River 230kV line, Red Oak unit 1 and 4 | 5 / 16                                          | Stable         |
| P4.26           | SLG @ South River Junction 230kV, normal clear loss of South River Junction – Red Oak – Raritan River 230kV line, Red Oak unit 2 and 3, SB @ South River Junction, delayed clear loss of South River Junction – Parlin 230kV                                             | 5 / 16                                          | Stable         |
| P4.27           | SLG @ South River Junction 230kV, normal clear loss of South River Junction – Atlantic 230kV, SB @ South River Junction, delayed clear loss of South River Junction – Parlin 230kV                                                                                       | 5 / 16                                          | Stable         |
| P4.28           | SLG @ South River Junction 230kV, normal clear loss of South River Junction – Atlantic 230kV, SB @ South River Junction, delayed clear loss of South River Junction – South River 230kV                                                                                  | 5 / 16                                          | Stable         |
| P4.29           | SLG @ South River Junction 230kV, normal clear loss of South River Junction – AC1-029 POI 230kV, SB @ South River Junction, delayed clear loss of South River Junction – South River 230kV, loss of AC1-029 and South River generators                                   | 5 / 16                                          | Stable         |
| P4.30           | SLG @ South River Junction 230kV, normal clear loss of South River Junction – AC1-029 POI 230kV, SB @ South River Junction, delayed clear loss of South River Junction – Red Oak – Raritan River 230kV line, Red Oak unit 1 and 4                                        | 5 / 16                                          | Stable         |
| P4.31           | SLG @ Parlin 230kV, normal clear loss of Parlin Load (Dupont), Parlin unit 1 and 2, SB @ Parlin 230kV, delayed clear loss of Parlin – South River Junction 230kV                                                                                                         | 5 / 16                                          | Stable         |
| P4.32           | SLG @ Parlin 230kV, normal clear loss of Parlin Load (Dupont), Parlin unit 1 and 2, SB @ Parlin 230kV, delayed clear loss of Parlin – Williams Gas – Freneau 230kV                                                                                                       | 5 / 16                                          | Stable         |
| P4.33           | SLG @ Parlin 230kV, normal clear loss of Parlin unit 3 and 4, SB @ Parlin 230kV, delayed clear loss of Parlin – South River Junction 230kV                                                                                                                               | 5 / 16                                          | Stable         |
| P4.34           | SLG @ Parlin 230kV, normal clear loss of Parlin unit 3 and 4, SB @ Parlin 230kV, delayed clear loss of Parlin – Williams Gas – Freneau 230kV                                                                                                                             | 5 / 16                                          | Stable         |
| P4.35           | SLG @ AC1-029 POI 230kV, normal clear loss of AC1-029 POI – South River 230 kV, SB @ AC1-029 POI, delayed clear loss of AC1-029, AC1-029 POI – South River Junction 230kV                                                                                                | 5 / 16                                          | Stable         |

| <b>Fault ID</b> | <b>Fault description</b>                                                                                                                                 | <b>Clearing Time Near &amp; Remote (Cycles)</b> | <b>Results</b> |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------|
| P4.36           | SLG @ South River 230kV, normal clear loss of South River – AC1-029 POI 230kV, SB @ South River, delayed clear loss of South River unit 1                | 5 / 16                                          | Stable         |
| P4.37           | SLG @ South River 230kV, normal clear loss of South River – AC1-029 POI 230kV, SB @ South River, delayed clear loss of South River unit 2 and 3          | 5 / 16                                          | Stable         |
| P4.38           | SLG @ South River 230kV, normal clear loss of South River – South River Junction 230kV, SB @ South River, delayed clear loss of South River unit 1       | 5 / 16                                          | Stable         |
| P4.39           | SLG @ South River 230kV, normal clear loss of South River – South River Junction 230kV, SB @ South River, delayed clear loss of South River unit 2 and 3 | 5 / 16                                          | Stable         |

#### **P5: SLG Fault with Delayed Clearing**

| <b>Fault ID</b> | <b>Fault description</b>                                                                                                               | <b>Clearing Time Near &amp; Remote (Cycles)</b> | <b>Results</b> |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------|
| P5.01           | SLG @ 80% of Raritan River – Kilmer – Lake Nelson 230kV ckt 1, delayed clear at Raritan River                                          | 5 / 34                                          | Stable         |
| P5.02           | SLG @ 80% of Lake Nelson – Kilmer – Raritan River 230kV ckt 1, delayed clear at Lake Nelson                                            | 5 / 34                                          | Stable         |
| P5.03           | SLG @ 80% of Raritan River – Kilmer – Lake Nelson 230kV ckt 2, delayed clear at Raritan River                                          | 5 / 34                                          | Stable         |
| P5.04           | SLG @ 80% of Lake Nelson – Kilmer – Raritan River 230kV ckt 2, delayed clear at Lake Nelson                                            | 5 / 34                                          | Stable         |
| P5.05           | SLG @ 80% of Raritan River – Red Oak – South River Junction 230kV, delayed clear at Raritan River, loss of Red Oak unit 2 and 3        | 5*                                              | Stable         |
| P5.06           | SLG @ 80% of South River Junction – Red Oak – Raritan River 230kV, delayed clear at South River Junction, loss of Red Oak unit 2 and 3 | 5*                                              | Stable         |
| P5.07           | SLG @ 80% of Raritan River – Red Oak – South River Junction 230kV, delayed clear at Raritan River, loss of Red Oak unit 1 and 4        | 5*                                              | Stable         |

| <b>Fault ID</b> | <b>Fault description</b>                                                                                                               | <b>Clearing Time Near &amp; Remote (Cycles)</b> | <b>Results</b> |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------|
| P5.08           | SLG @ 80% of South River Junction – Red Oak – Raritan River 230kV, delayed clear at South River Junction, loss of Red Oak unit 1 and 4 | 5*                                              | Stable         |
| P5.09           | SLG @ 80% of Raritan River – NJ Steel – Werner 230kV, delayed clear at Raritan River                                                   | 5 / 34                                          | Stable         |
| P5.10           | SLG @ 80% of Werner – NJ Steel – Raritan River 230kV, delayed clear at Werner                                                          | 5 / 34                                          | Stable         |
| P5.11           | SLG @ 80% of South River Junction – Atlantic 230kV, delayed clear at South River Junction                                              | 5 / 34                                          | Stable         |
| P5.12           | SLG @ 80% of Atlantic – South River Junction 230kV, delayed clear at Atlantic                                                          | 5 / 34                                          | Stable         |
| P5.13           | SLG @ 80% of South River Junction – Parlin 230kV, delayed clear at South River Junction                                                | 5*                                              | Stable         |
| P5.14           | SLG @ 80% of Parlin – South River Junction 230kV, delayed clear at Atlantic                                                            | 5*                                              | Stable         |
| P5.15           | SLG @ 80% of Parlin – Williams – Freneau 230kV, delayed clear at Parlin                                                                | 5*                                              | Stable         |
| P5.16           | SLG @ 80% of Freneau – Williams – Parlin 230kV, delayed clear at Parlin                                                                | 5*                                              | Stable         |

**\*Dual communication scheme exists**

## **P7: SLG Fault with Common Structure Failure**

| <b>Fault ID</b> | <b>Fault description</b>                                                                | <b>Clearing Time (Cycles)</b> | <b>Results</b> |
|-----------------|-----------------------------------------------------------------------------------------|-------------------------------|----------------|
| P7.01           | CONTINGENCY 'JC-P7-1-JCC-230-18T' /* LAKE NELSON - GREEN BROCK - GILLETTE I1023 & W1037 | 5                             | Stable         |
| P7.02           | CONTINGENCY 'JC-P7-1-JCC-230-3' /* PARLIN-RED OAK-RAR R & S.RIVER-RED OAK-RAR RIVER     | 5                             | Stable         |
| P7.03           | CONTINGENCY 'JC-P7-1-JCC-230-4' /* R11B-ATLANTIC (P1030) & PARLIN-FRENEAU (K1025)       | 5                             | Stable         |
| P7.04           | CONTINGENCY 'JC-P7-1-JCC-230-5A' /* S.RVR-ATLANTIC & FRENEAU-ATLANTIC 230 KV            | 5                             | Stable         |

| <b>Fault ID</b> | <b>Fault description</b>                                                                             | <b>Clearing Time (Cycles)</b> | <b>Results</b> |
|-----------------|------------------------------------------------------------------------------------------------------|-------------------------------|----------------|
| P7.05           | CONTINGENCY 'JC-P7-1-JCC-230-6B' /* DEEP RUN-RAR RVR B2 & C3 115 KV FUTURE DEEP RUN BREAKERS         | 5                             | Stable         |
| P7.06           | CONTINGENCY 'JC-P7-1-JCC-230-9AT' /* RAR R-NEJESCO-WERNER 230 KV & RAR R-WERNER 115 KV               | 5                             | Stable         |
| P7.07           | CONTINGENCY 'PS_P7-1_I1023+GBK-LN_LT' /* LAKE NELSON - GILLETE BRIDGEWATER & LAKE NELSON -GREENBROOK | 5                             | Stable         |
| P7.08           | CONTINGENCY 'PS_P7-1_I1023+W1037' /* LAKE NELSON - GILLETE BRIDGEWATER & GILLETE -GREENBROOK         | 5                             | Stable         |